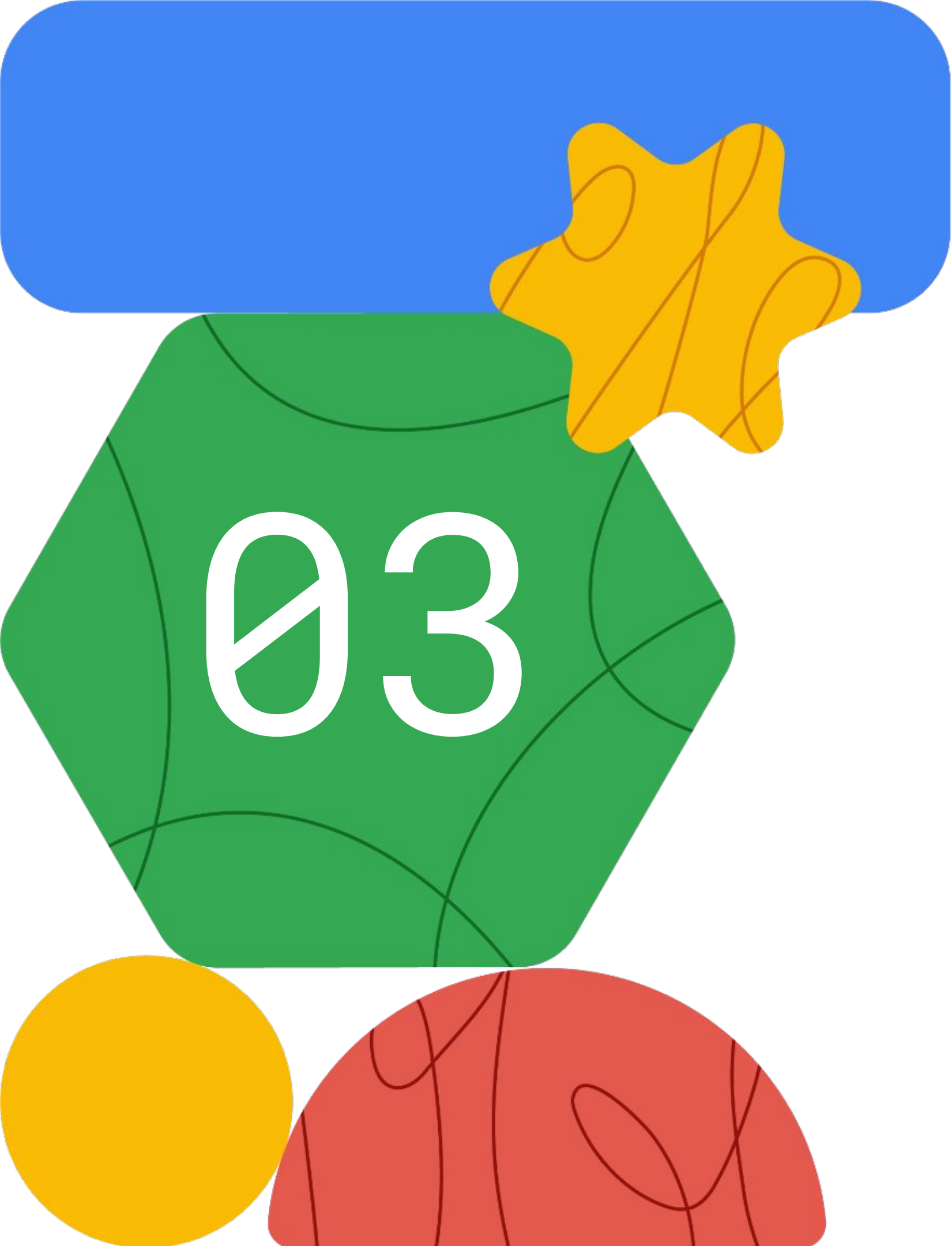


Video Info

File name: P-CDL-I-M3-1-Innovating with Google Cloud Artificial Intelligence

Presenter:



03

Innovating with Google Cloud Artificial Intelligence

Module 3

Innovating with Google Cloud Artificial Intelligence



Lessons

- 01 AI and ML fundamentals
- 02 Google Cloud's AI and ML solutions

Module 3

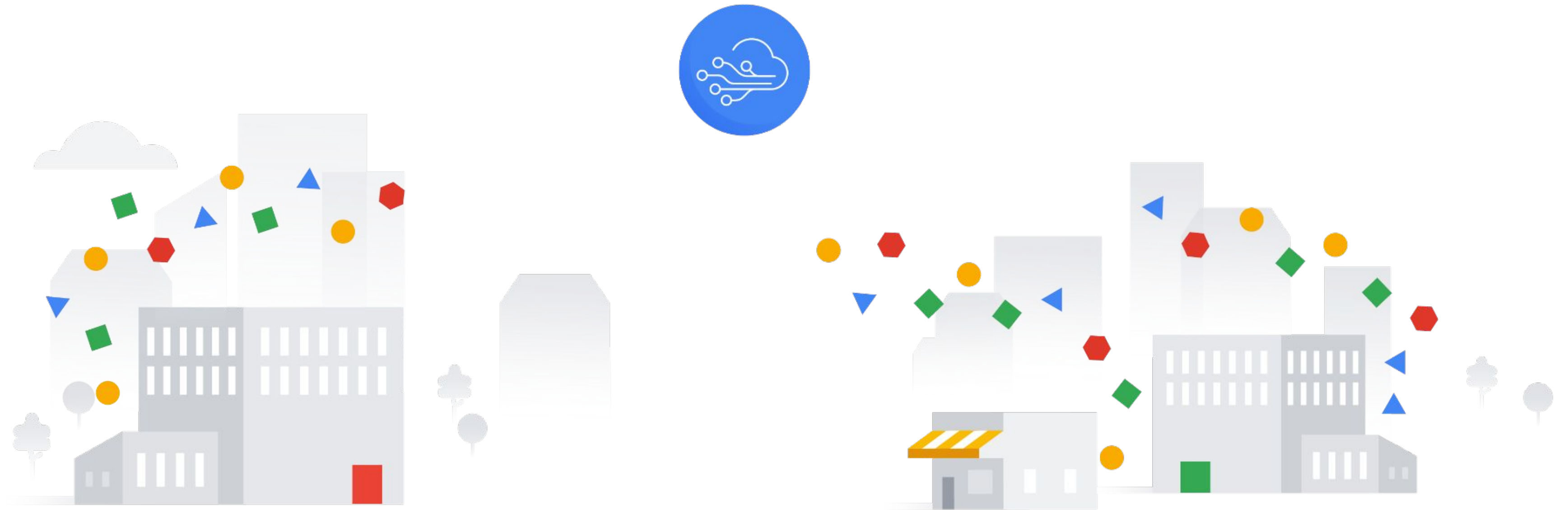
Innovating with Google Cloud Artificial Intelligence



Lessons

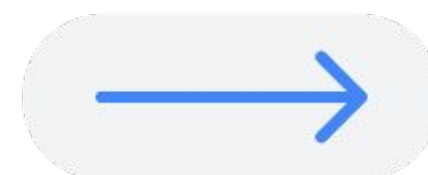
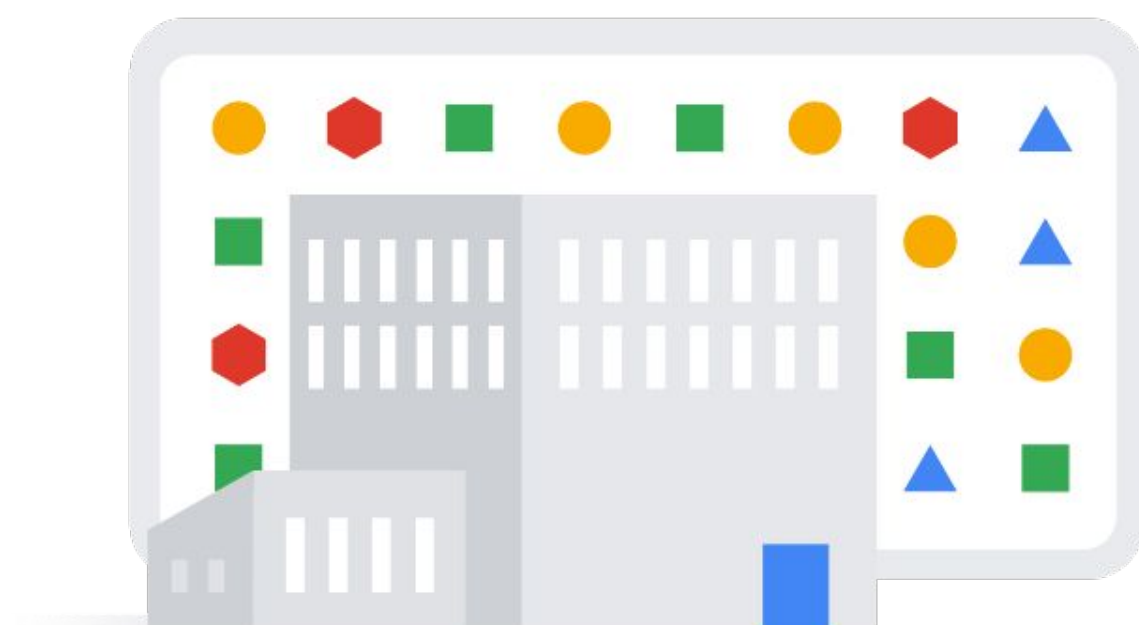
- 01 AI and ML fundamentals
- 02 Google Cloud's AI and ML solutions

Digital transformation brings lots of data



Data can be very laborious to collect, process, and analyze.

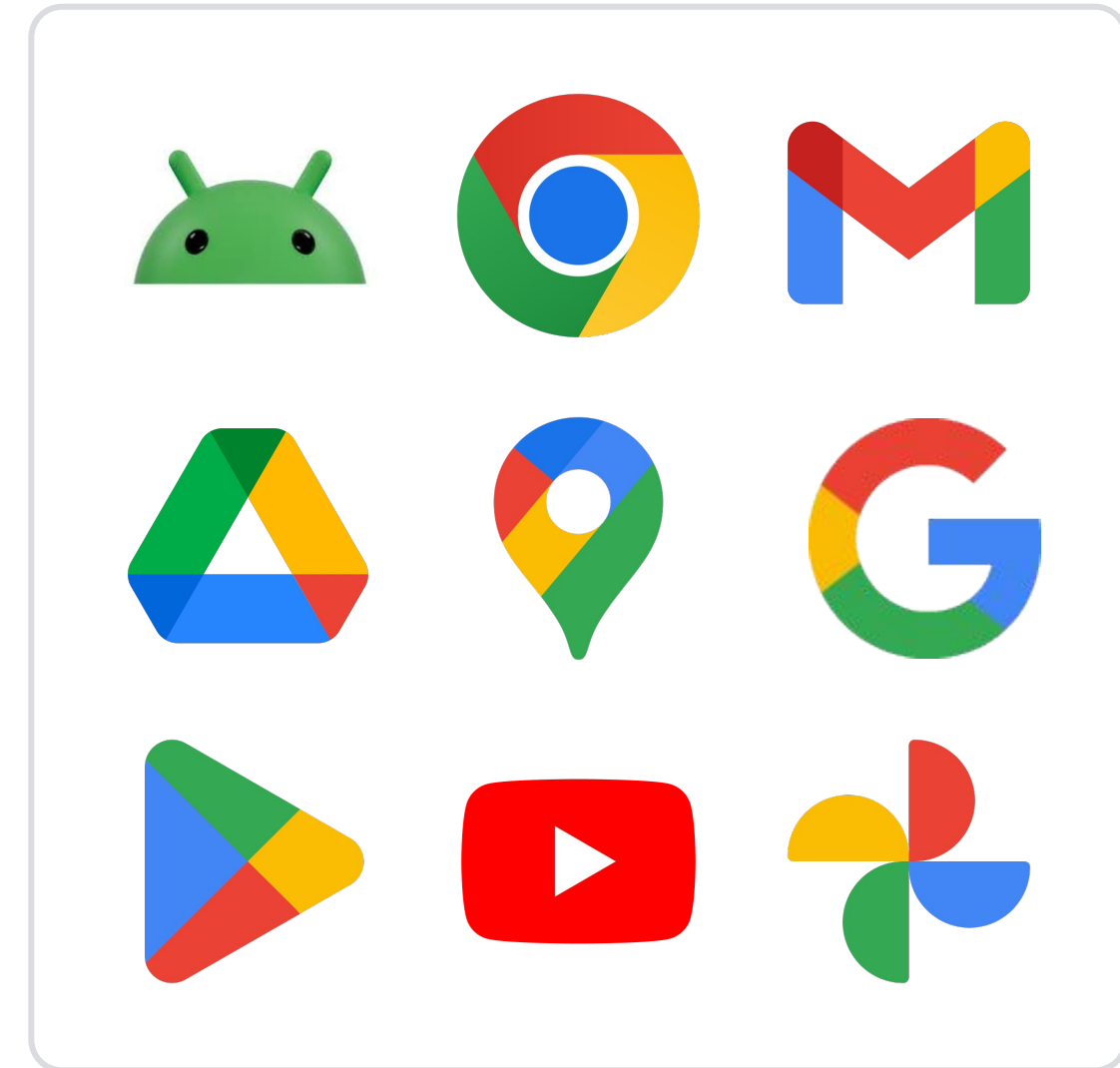
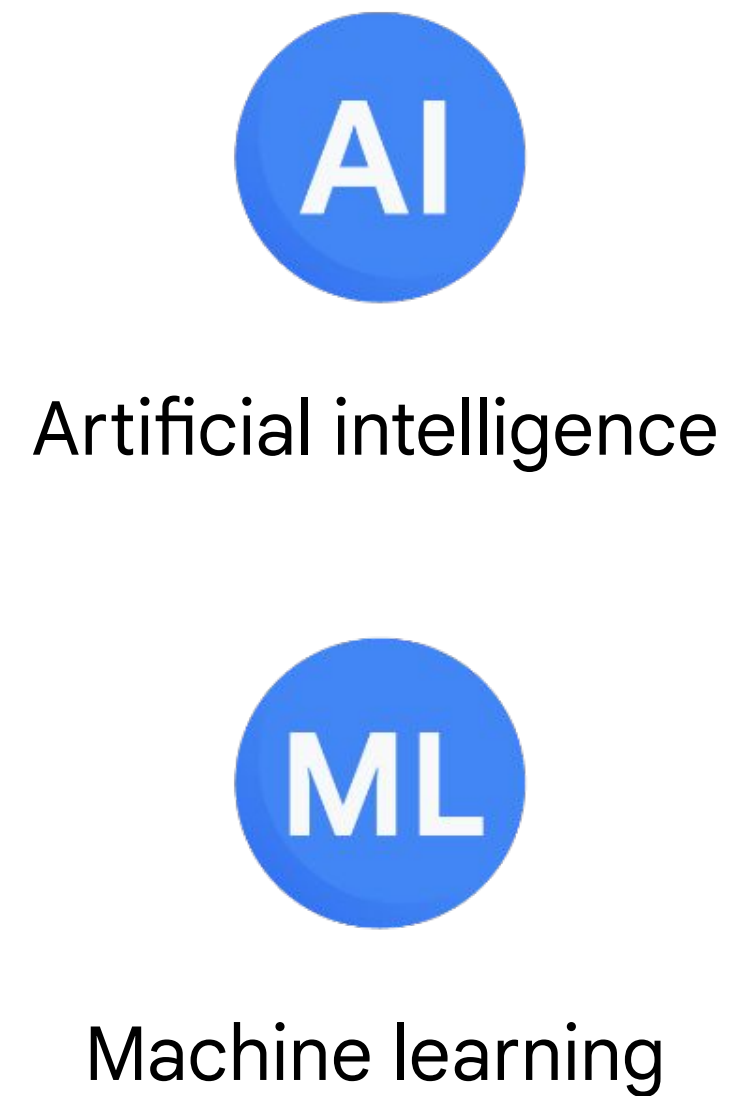
AI and ML can help



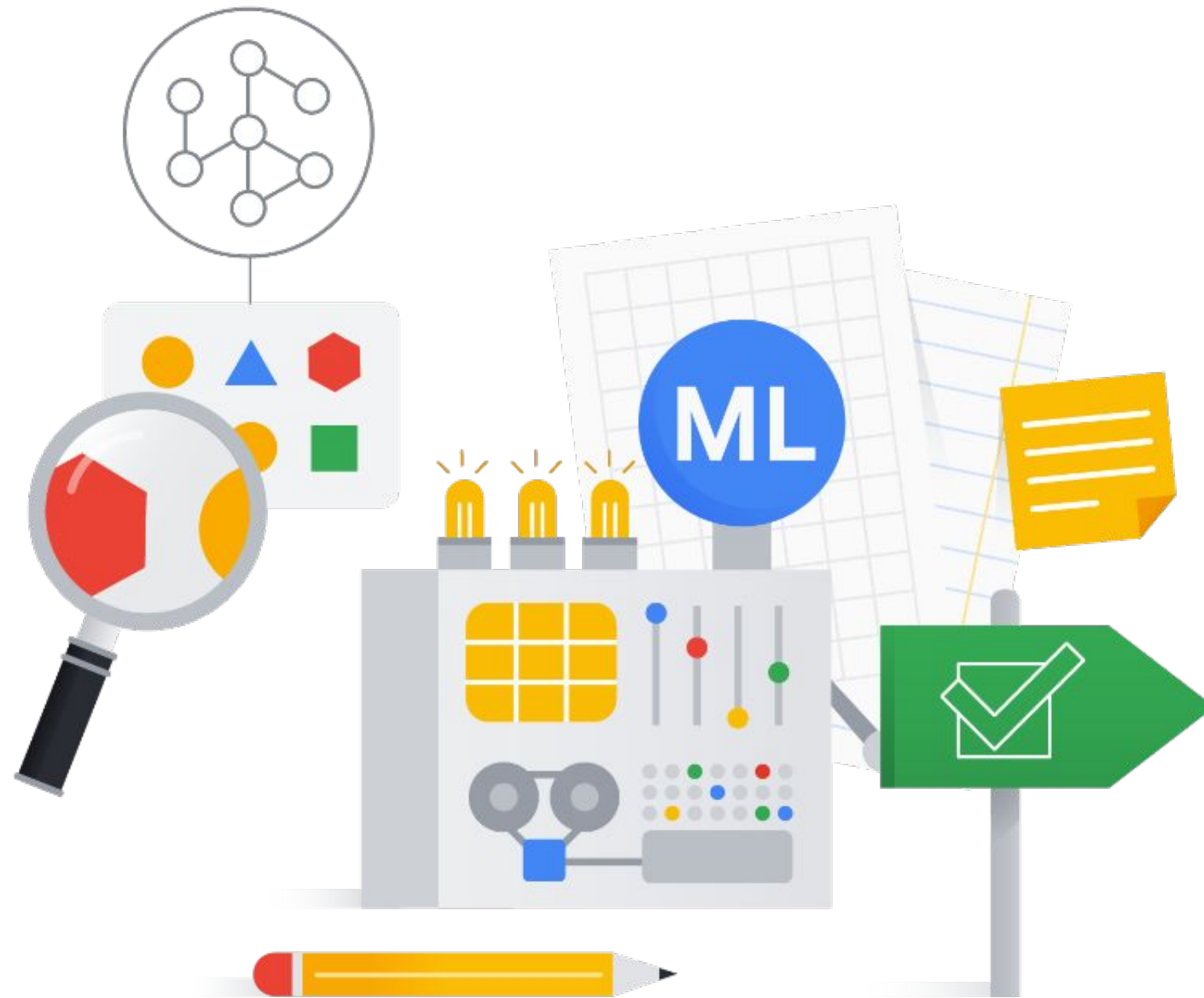
Artificial Intelligence

Machine Learning

AI and ML are integrated in Google's popular products



Machine learning is a subset of AI

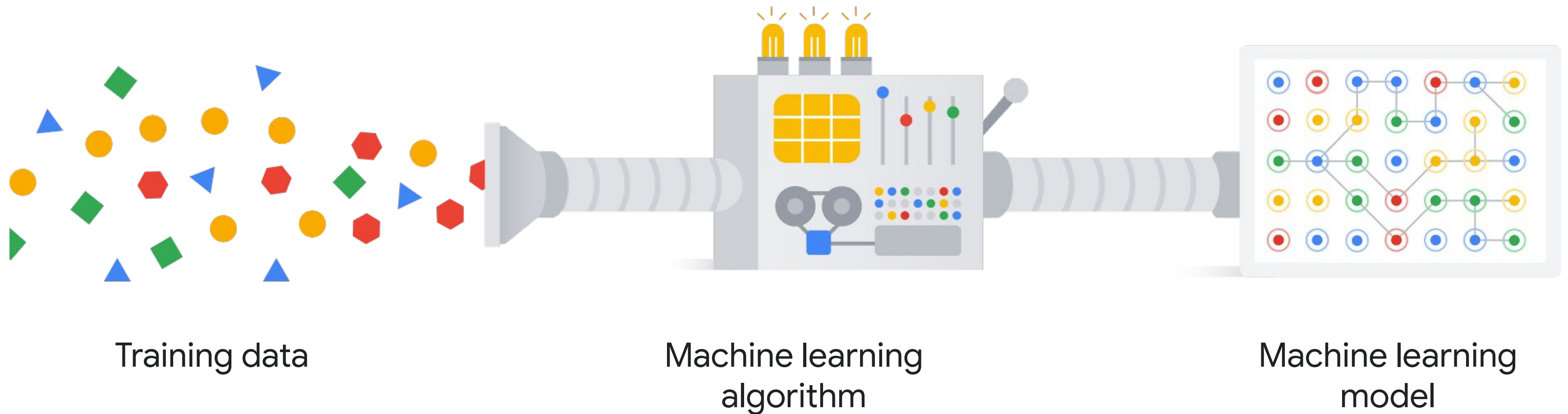


ML lets a machine learn from data without being explicitly programmed.

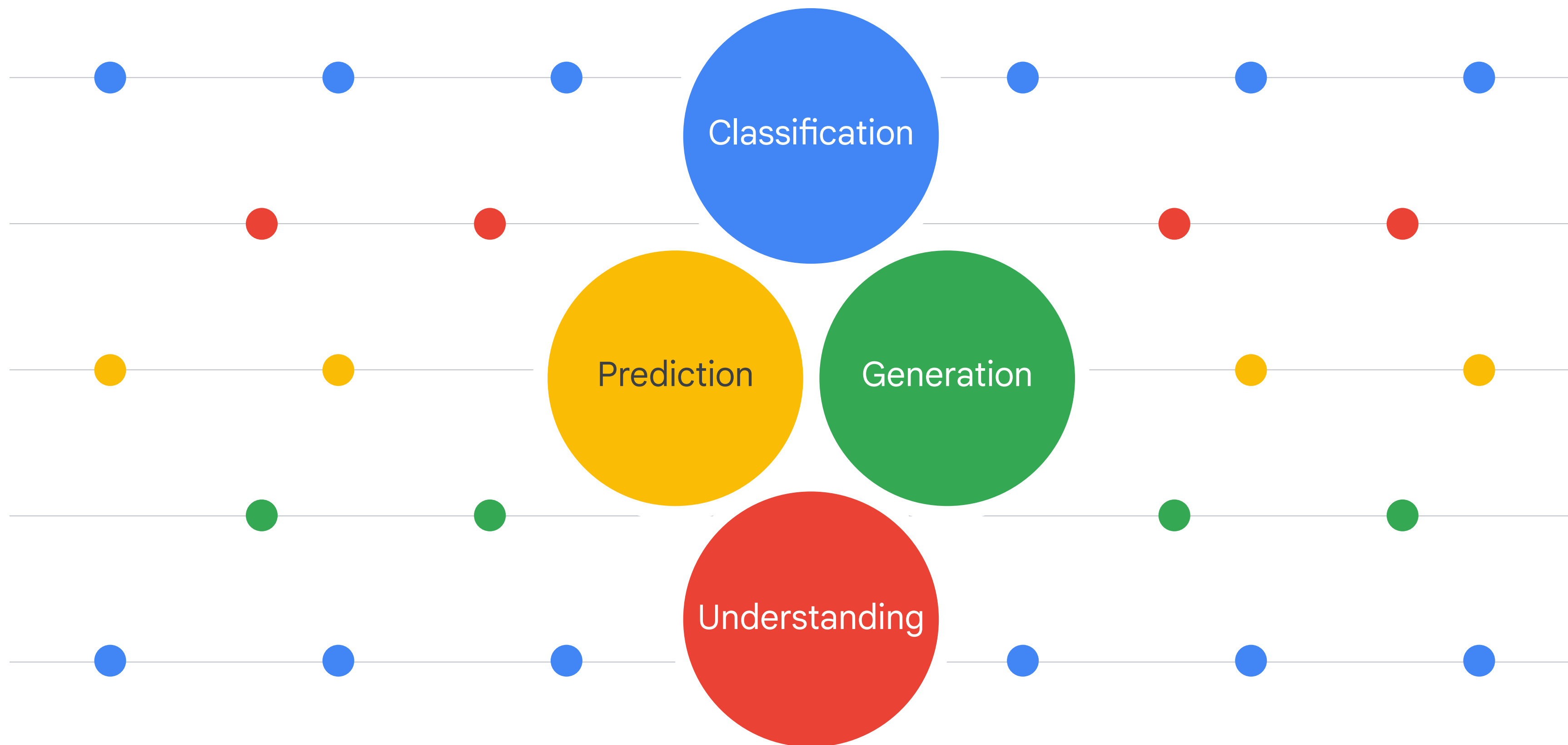
It relies on various **models** to:

- ✓ Analyze large amounts of data
- ✓ Learn from the insights
- ✓ Make predictions and informed decisions

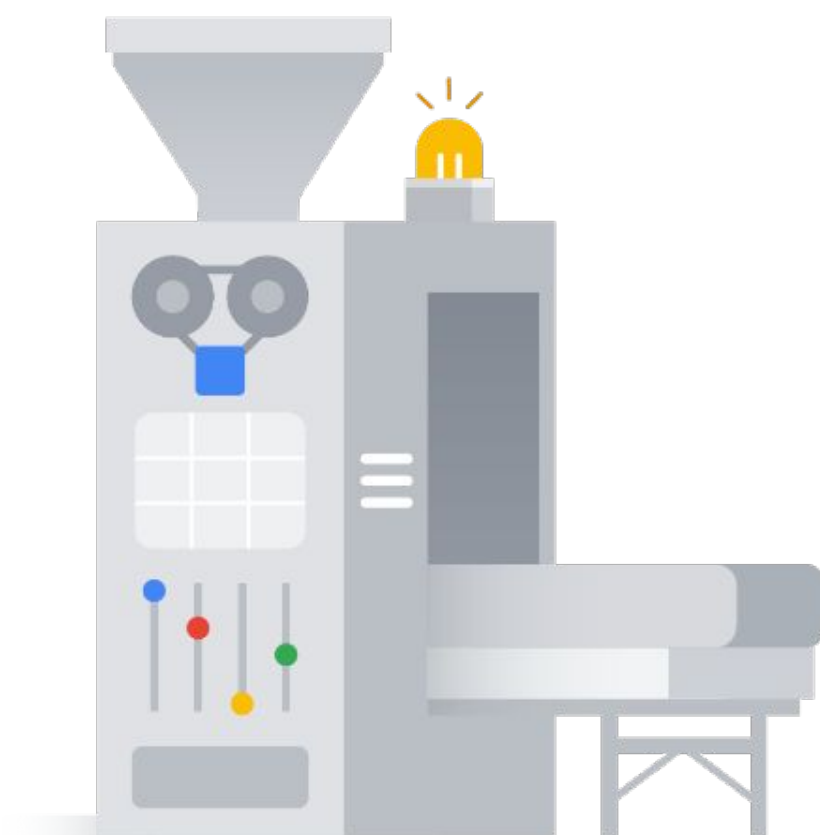
ML algorithms improve performance over time as they are trained or exposed to more data



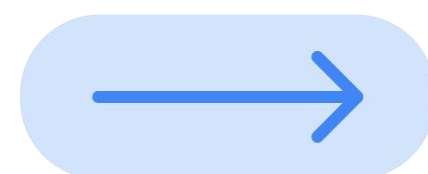
AI is evolving from analysis to creation



Generative AI

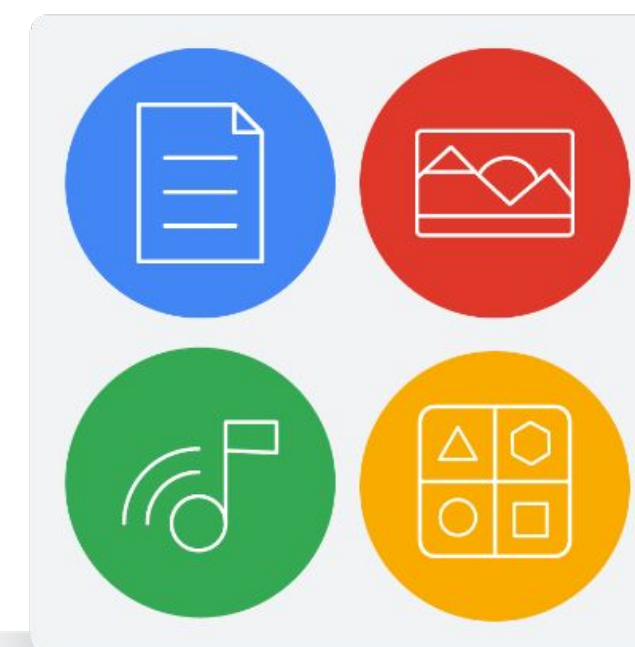


A type of artificial intelligence that can produce new content



Text

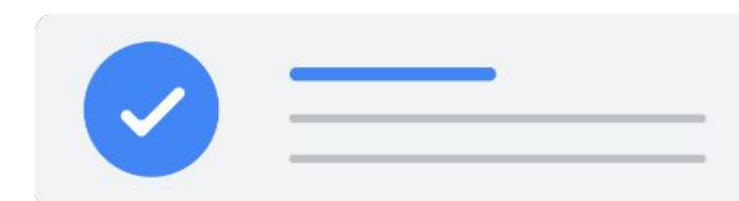
Audio



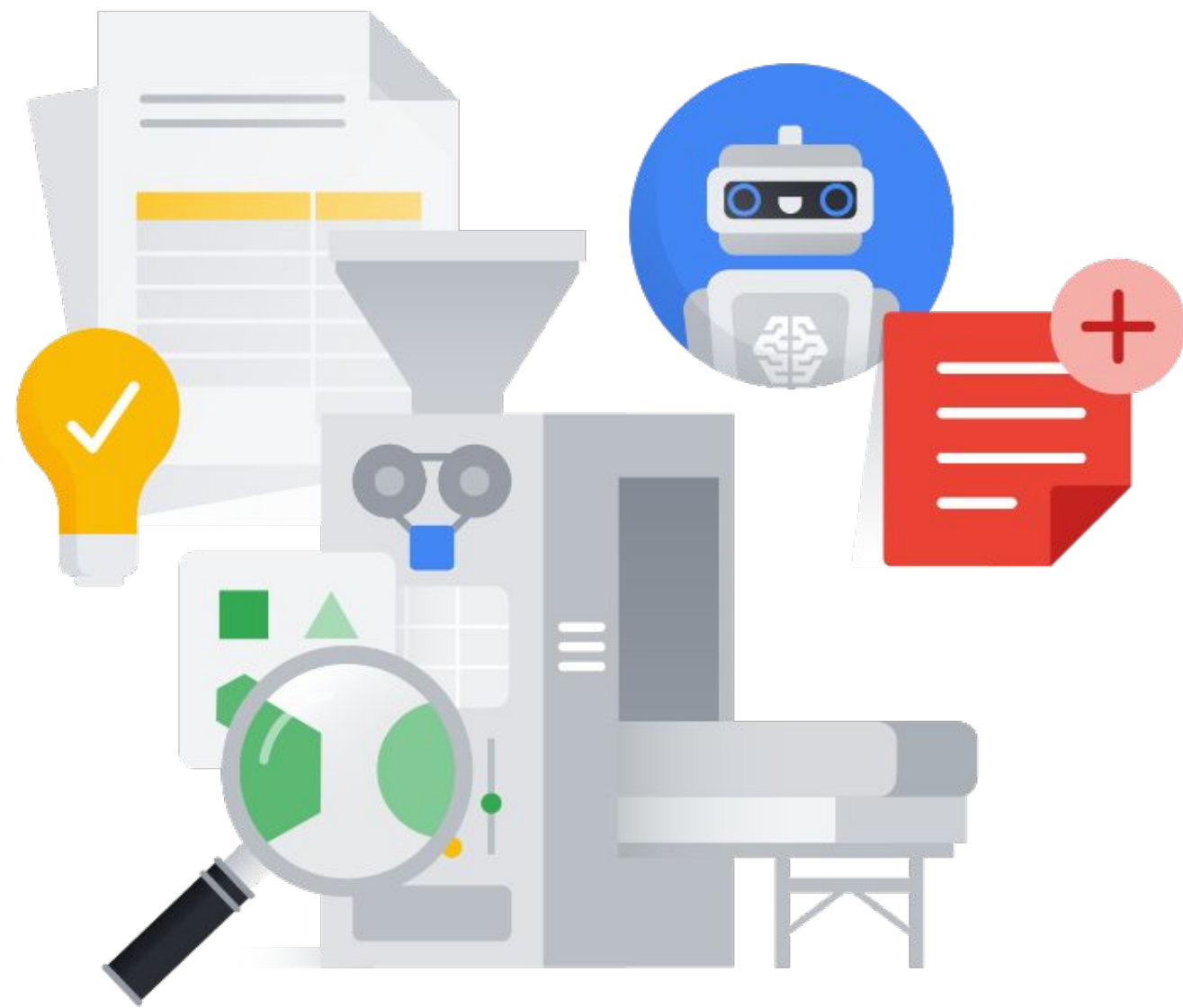
Images

Synthetic data

Google Workspace



Generative AI use cases



Generative AI can be used in:

- ✓ Conversational bots
- ✓ Content generation
- ✓ Document synthesis
- ✓ Product discovery

Quiz

Question

What is generative AI?

- A. A type of artificial intelligence that can understand and respond to human emotions
- B. A type of artificial intelligence that can produce new content, including text, images, and audio
- C. A type of artificial intelligence that can make decisions and take actions
- D. A type of artificial intelligence that can create and sustain its own consciousness

Quiz

Answer

What is generative AI?

- A. A type of artificial intelligence that can understand and respond to human emotions
- B. A type of artificial intelligence that can produce new content, including text, images, and audio
- C. A type of artificial intelligence that can make decisions and take actions
- D. A type of artificial intelligence that can create and sustain its own consciousness



Quiz

Question

Which of these is a generative AI use case?

- A. Self-driving cars
- B. Spam filtering
- C. Conversational bots
- D. Image recognition

Quiz

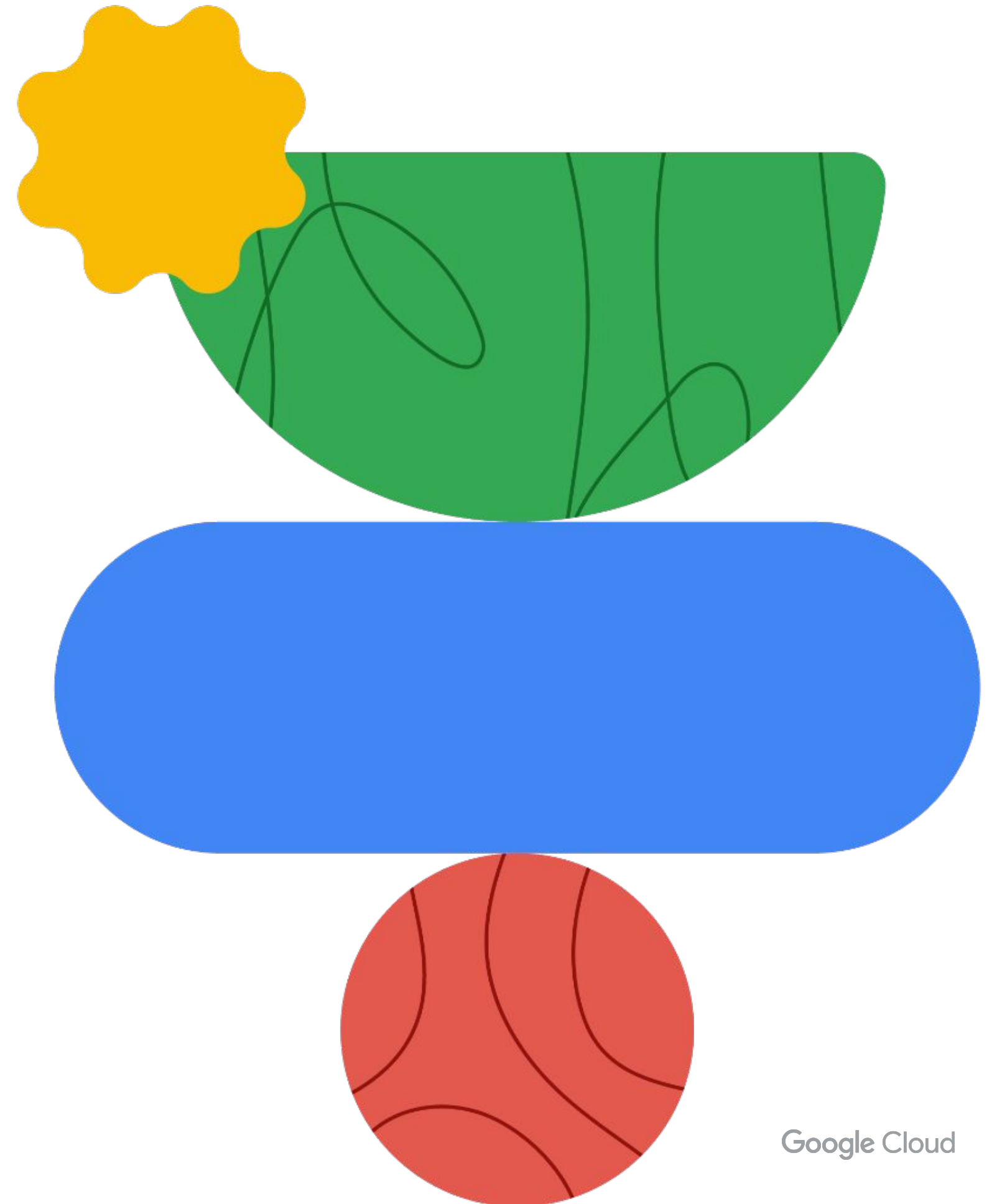
Answer

Which of these is a generative AI use case?

- A. Self-driving cars
- B. Spam filtering
- C. Conversational bots
- D. Image recognition



How AI and ML differ from data analytics and business intelligence



Backward-looking data



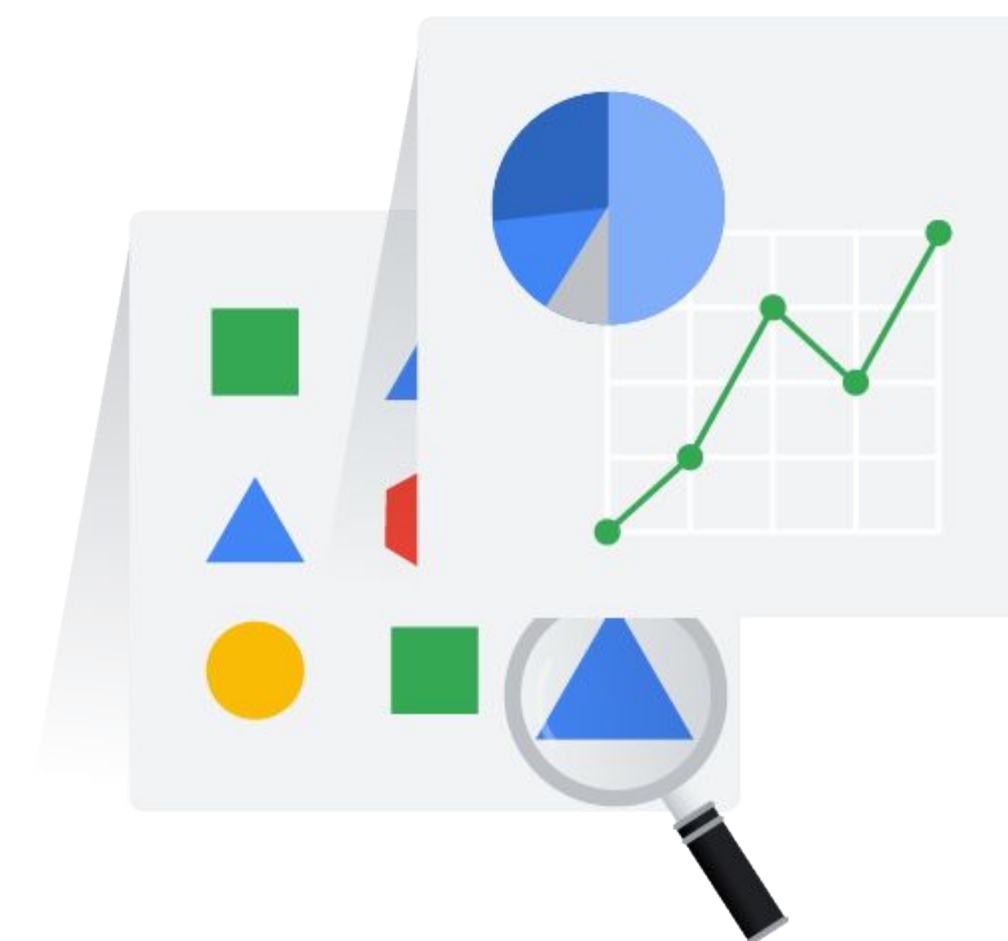
Historic data



Calculate
metrics



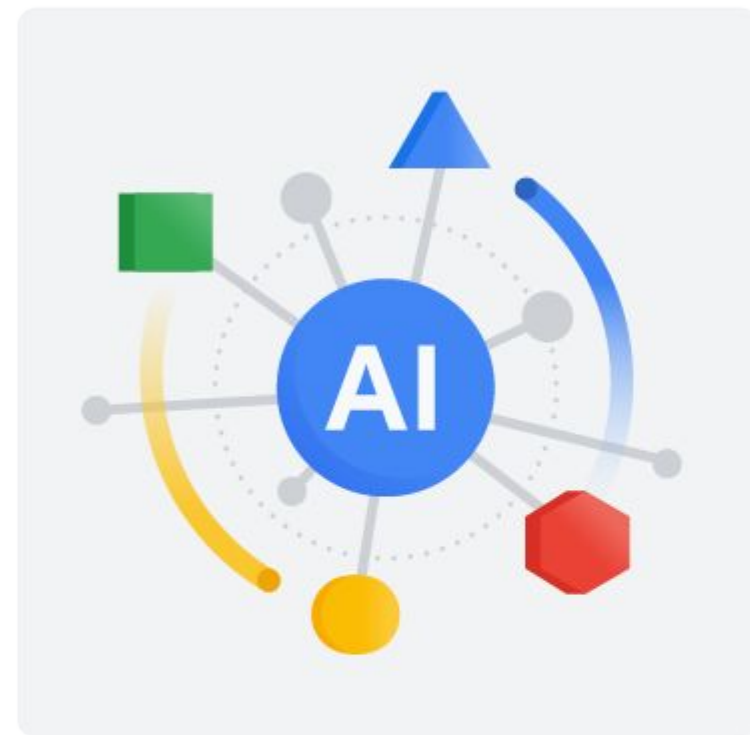
Identify
trends



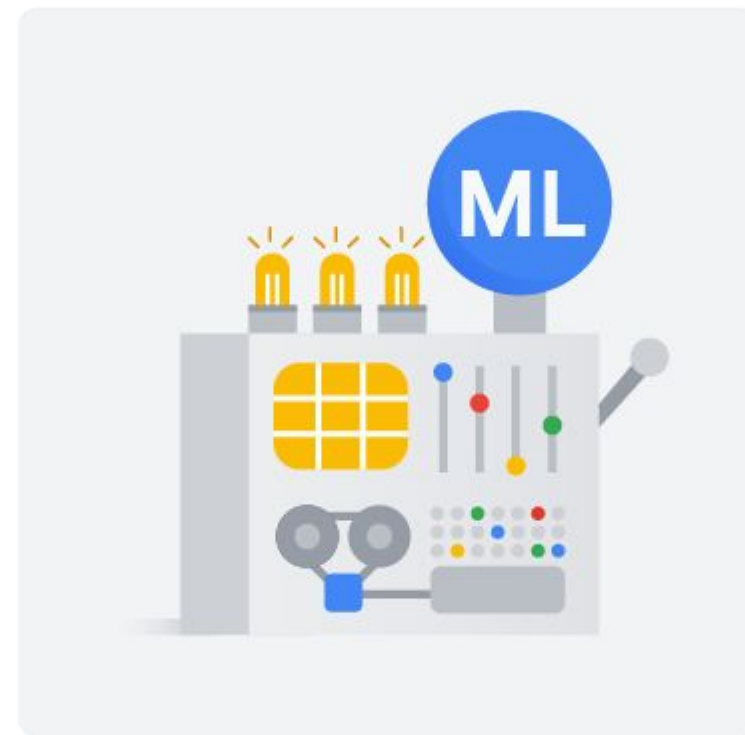
Data analysis

Business intelligence

To create value, you need to use data to make decisions for future business



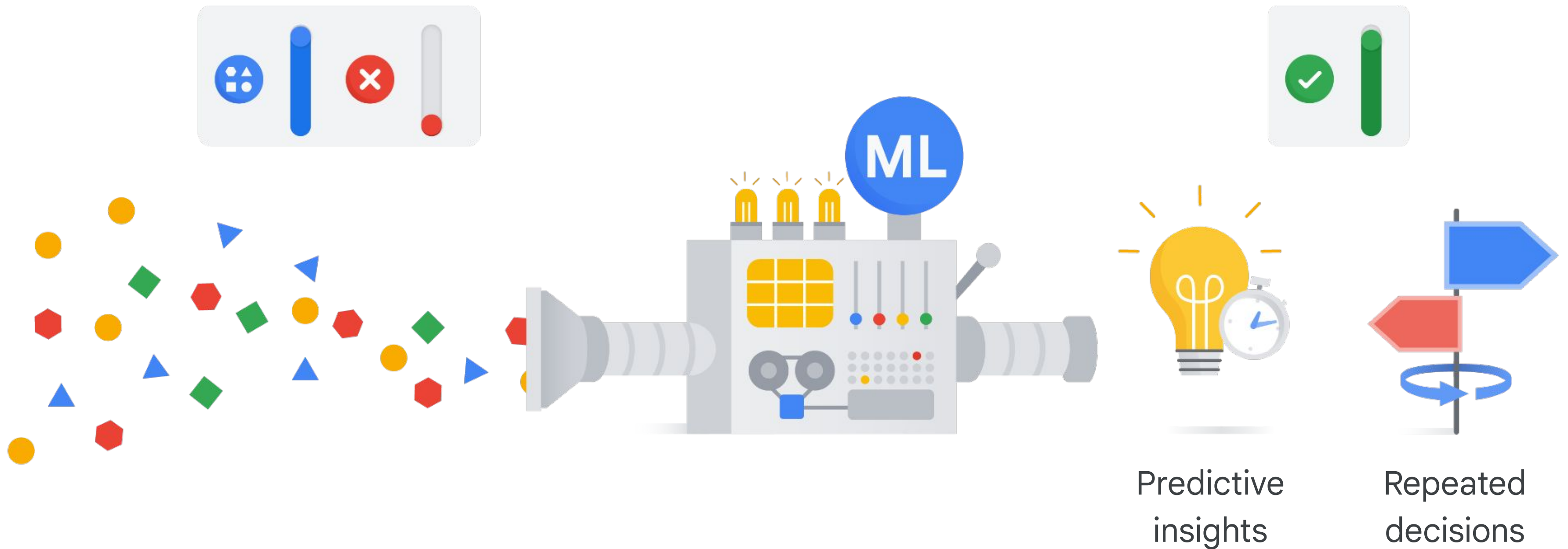
Artificial intelligence



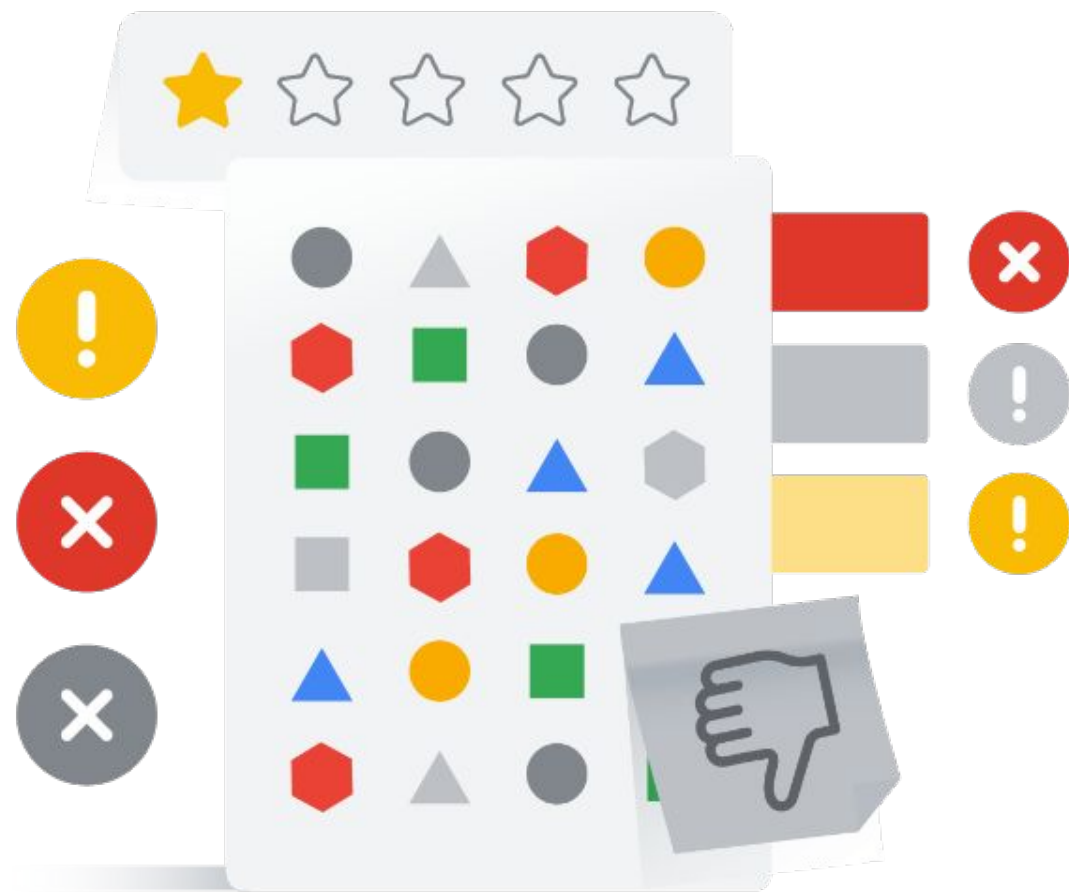
Machine learning



Data is used by ML models to derive predictive insights and make repeated decisions



An ML model can't make accurate predictions by learning from incorrect data

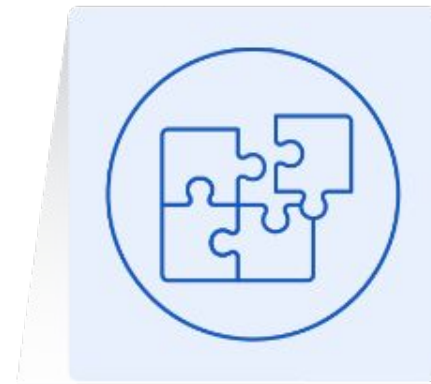


Low quality data is:

× Not aligned to the problem

× Biased in some way

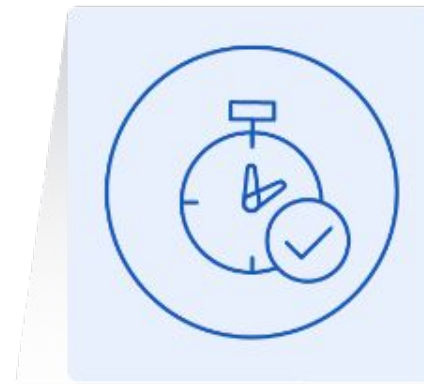
Data is evaluated against 6 dimensions to assess quality



Completeness



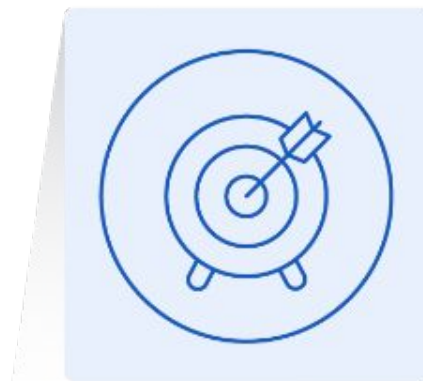
Uniqueness



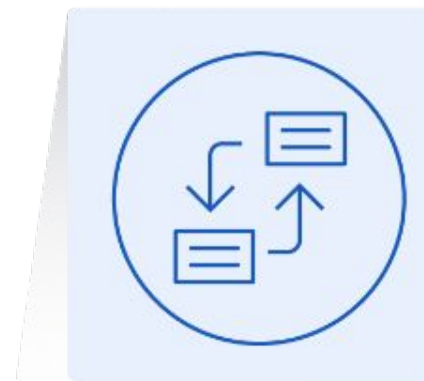
Timeliness



Validity



Accuracy



Consistency

It's your responsibility to provide a model with complete and correct data



Completeness



Uniqueness



Timeliness



Validity

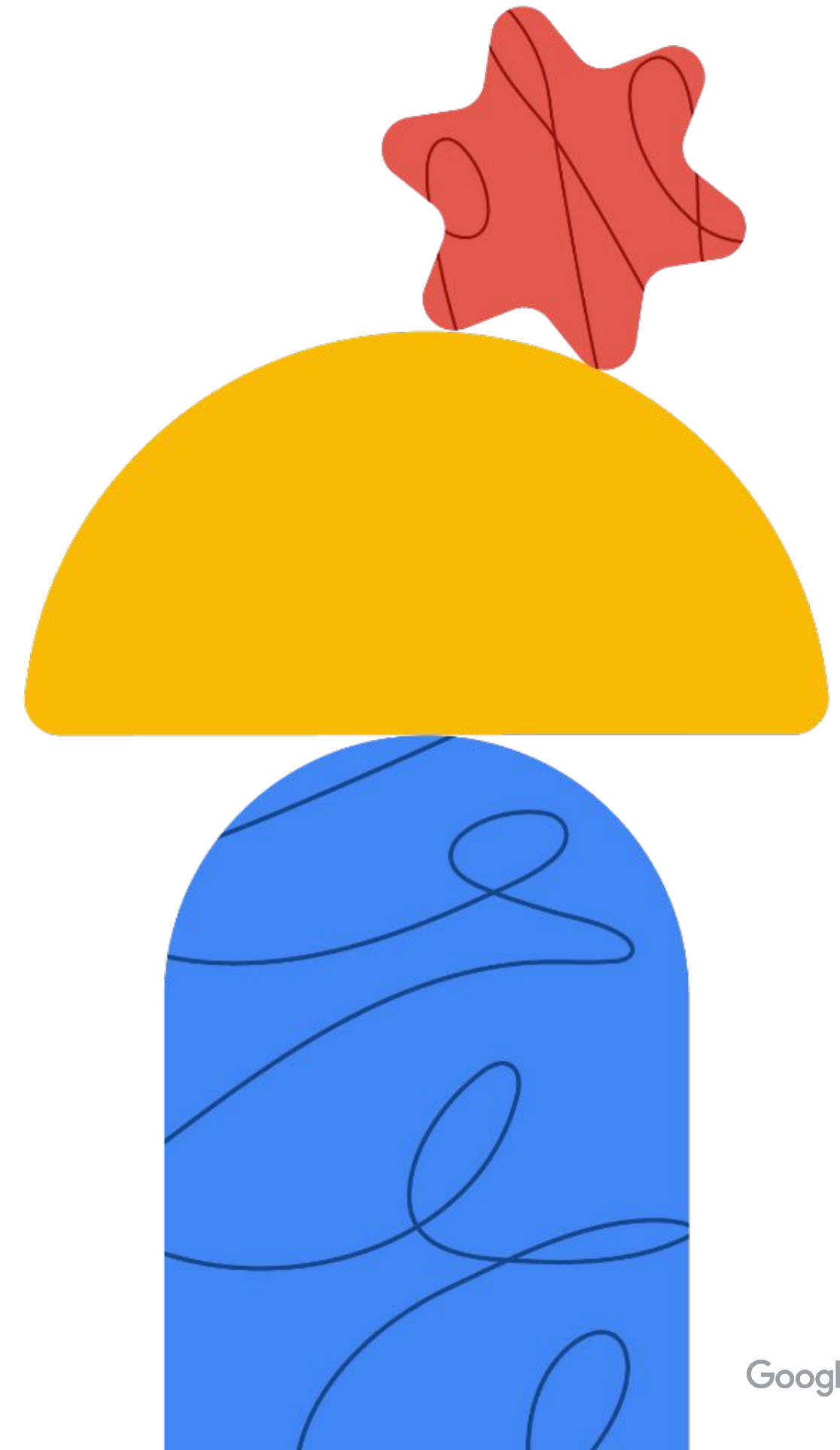


Accuracy

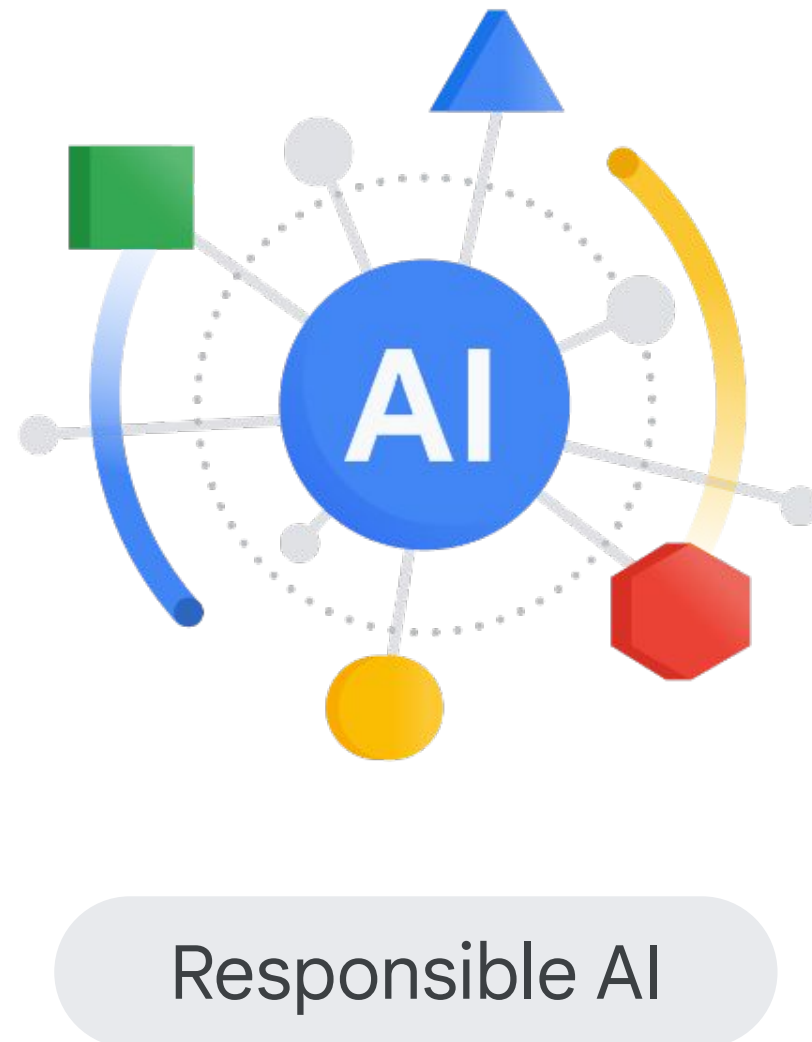


Consistency

The importance of responsible AI and explainable AI



It's critical that AI is used responsibly



Google has established:

- ✓ Principles that guide Google AI applications
- ✓ Best practices to share our work with communities outside of Google
- ✓ Programs to operationalize our efforts

Google's AI principles

AI should be socially beneficial.

AI should avoid creating or reinforcing unfair bias.

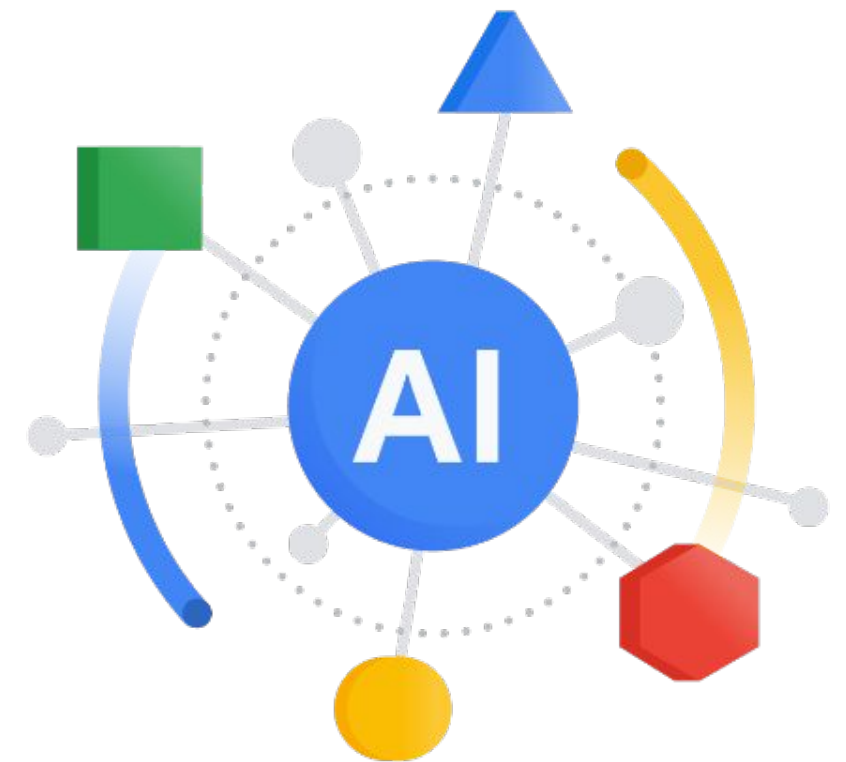
AI should be built and tested for safety.

AI should be accountable to people.

AI should incorporate privacy design principles.

AI should uphold high standards of scientific excellence.

AI should be made available for uses that accord with these principles.



When Google will not design or deploy AI



For technologies that cause or are likely to cause overall harm



For weapons or other technologies whose principal purpose or implementation is to cause or directly facilitate injury to people

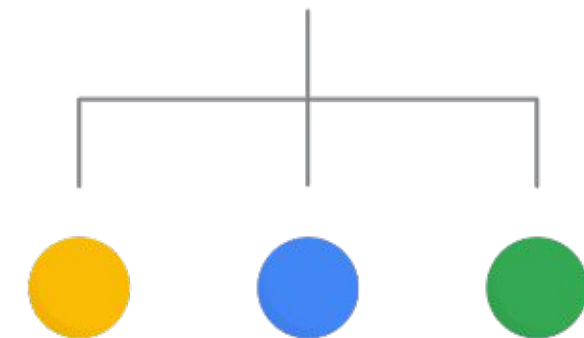
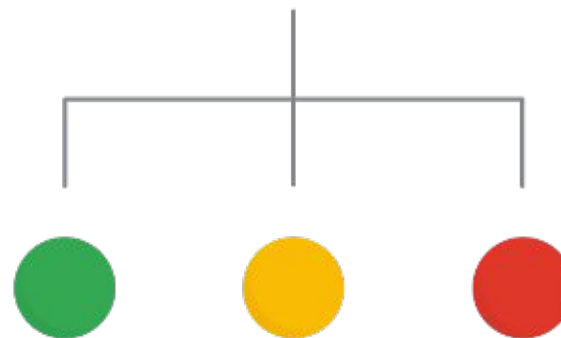
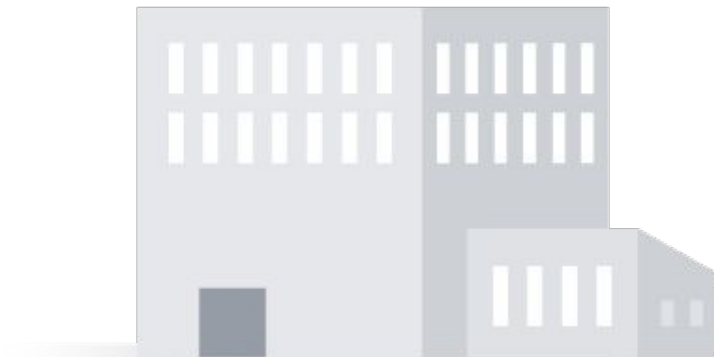
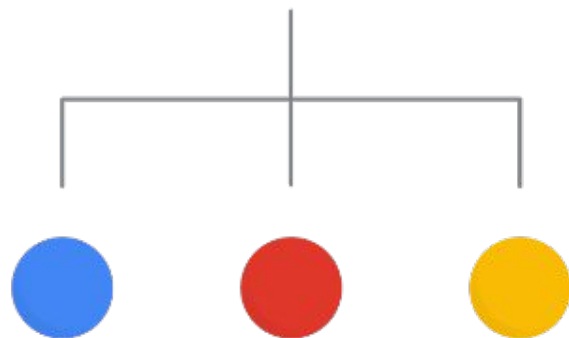
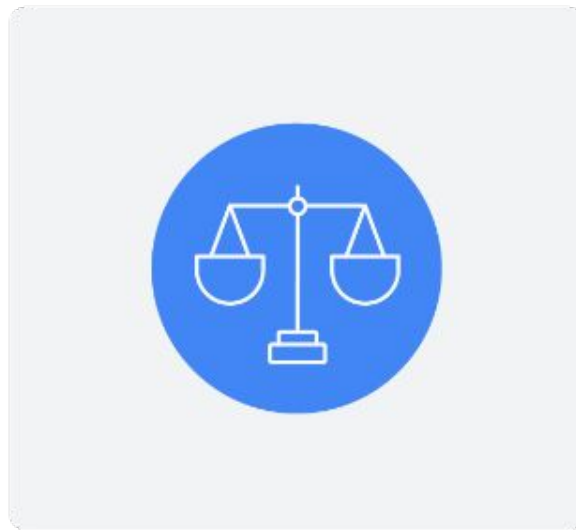


For technologies that gather or use information for surveillance, violating internationally accepted norms



For technologies whose purpose contravenes widely accepted principles of international law and human rights

Organizations should develop their own set of responsible AI principles



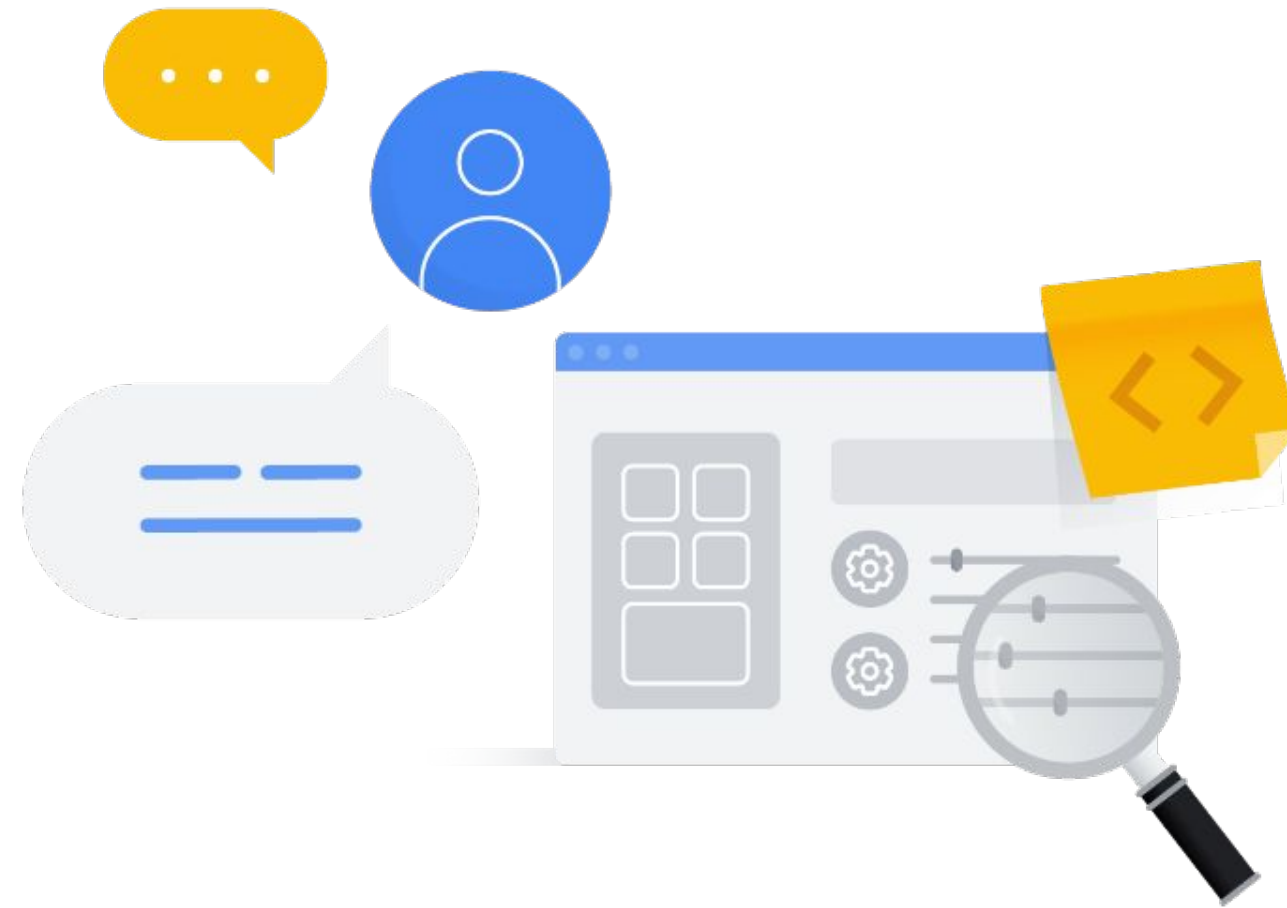
Other important considerations



Organizations should:

- ✓ Debug and improve ML model performance.
- ✓ Help others understand model behavior.
- ✓ Detect and resolve bias, drift, and other gaps.
- ✓ Have human-interpretable explanations of ML models to help grow end-user trust and improve transparency.

Explainable AI



Google Cloud's set of tools and frameworks to help you understand and interpret predictions made by your machine learning models

ML options on Google Cloud

01

BigQuery ML

Use SQL queries to create and execute machine learning models in BigQuery.

02

Pre-trained APIs

Use ML models that have already been built and trained by Google.

03

AutoML

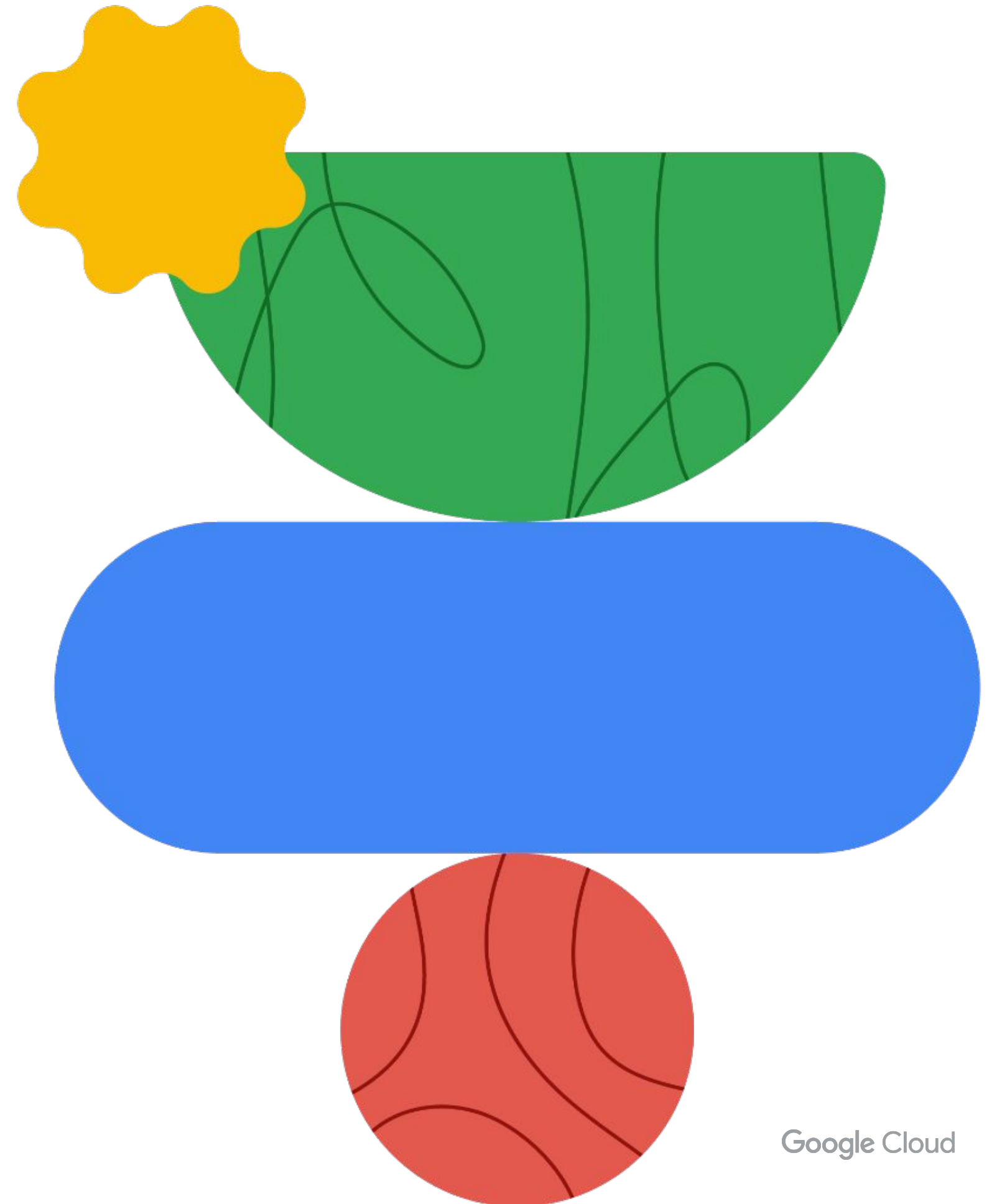
It's a no-code solution to build ML models on Vertex AI.

04

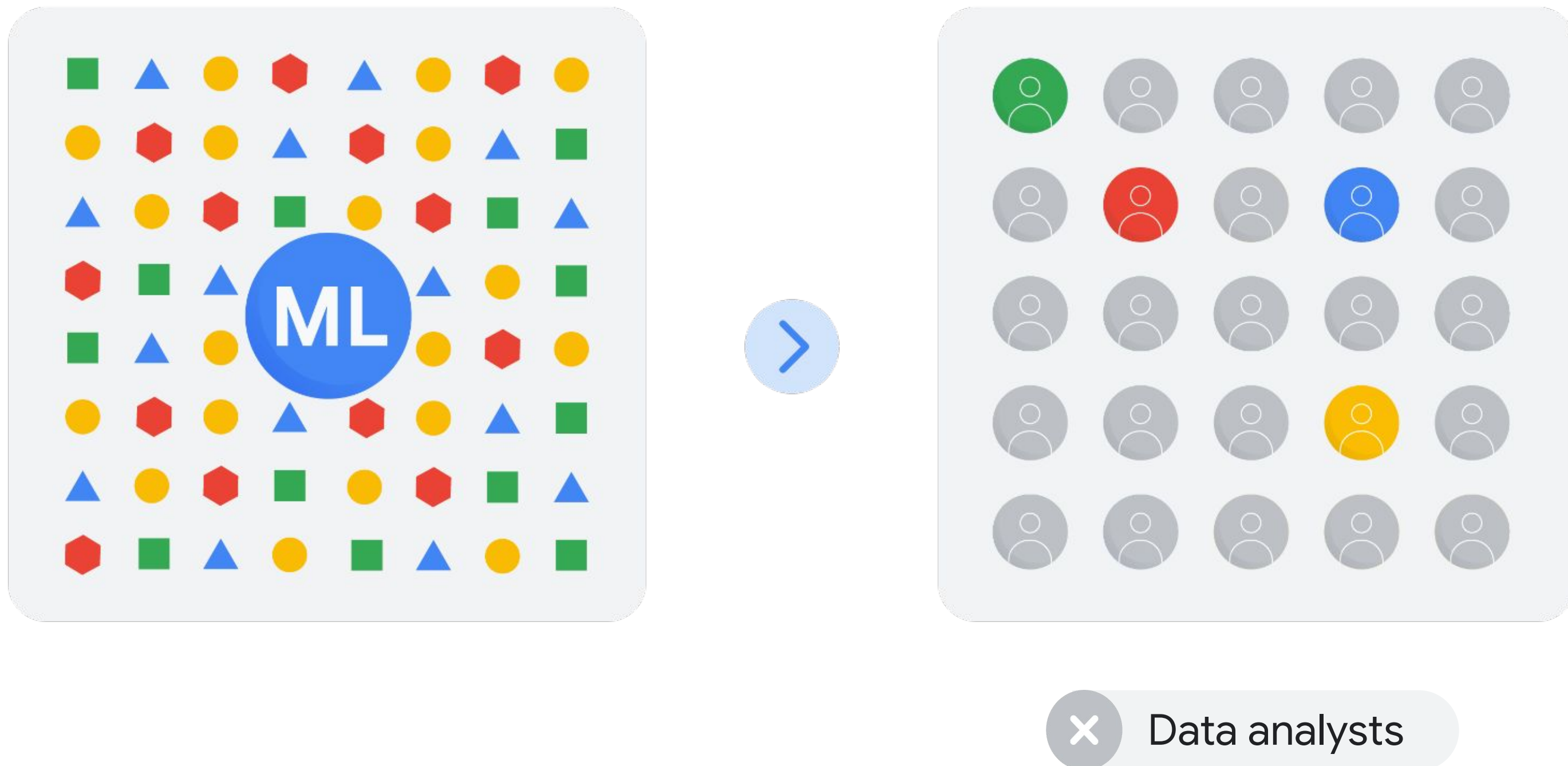
Custom training

Code your own ML environment to have the control over the ML pipeline.

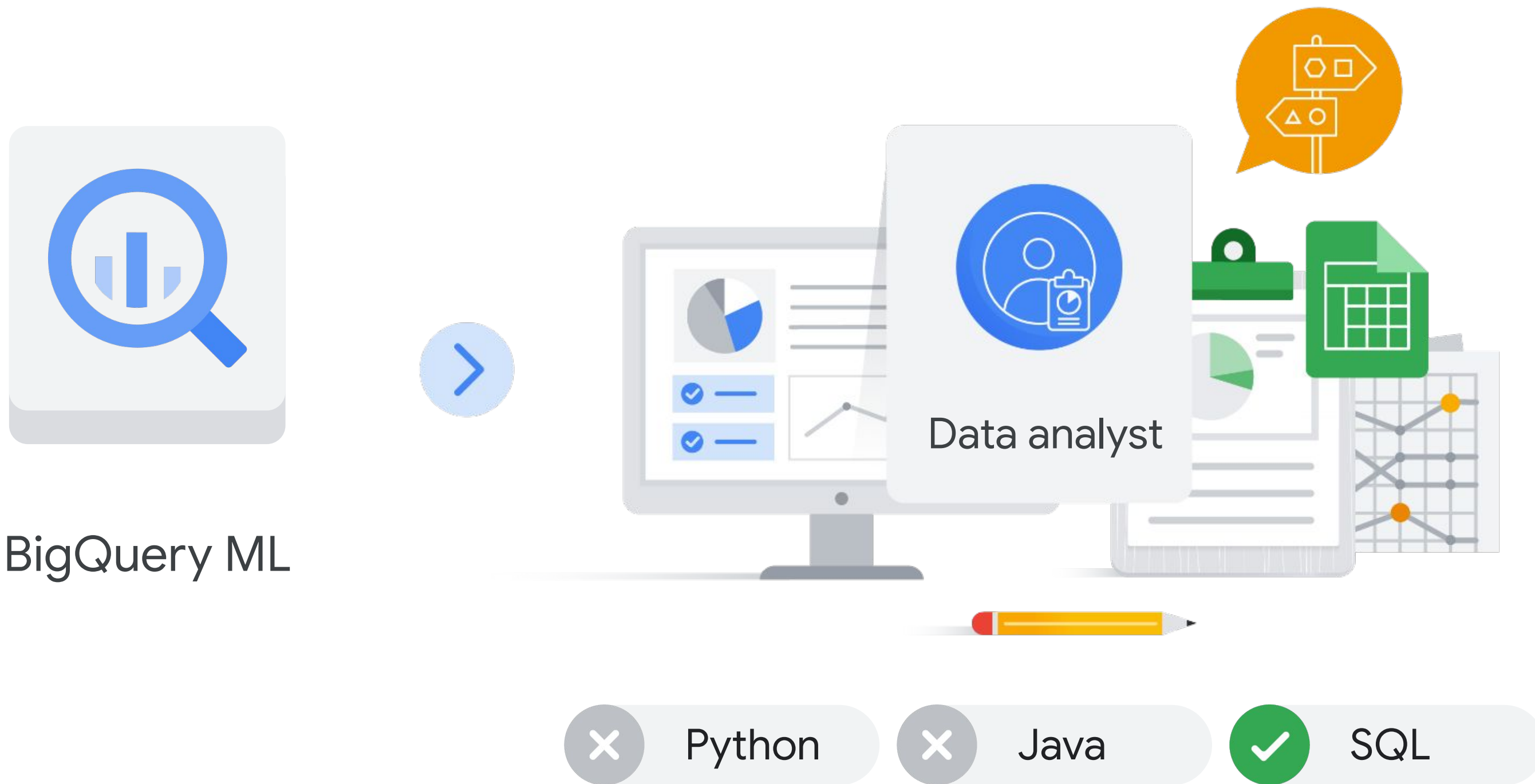
BigQuery ML



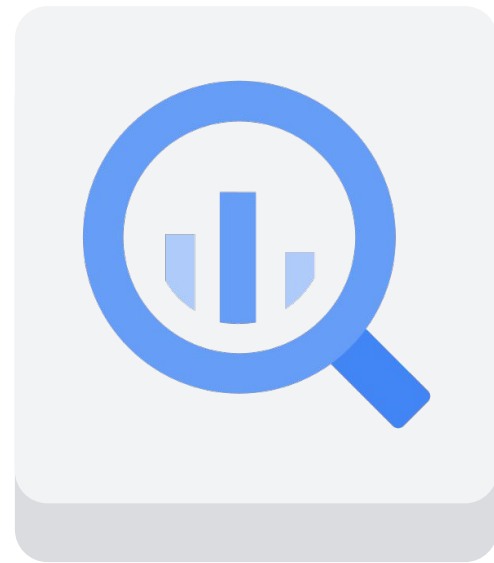
ML on large data sets requires extensive programming and knowledge of ML frameworks



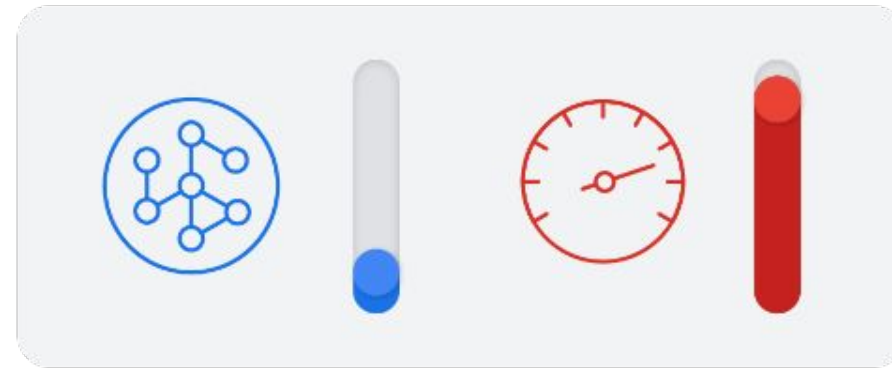
BigQuery ML empowers data analysts



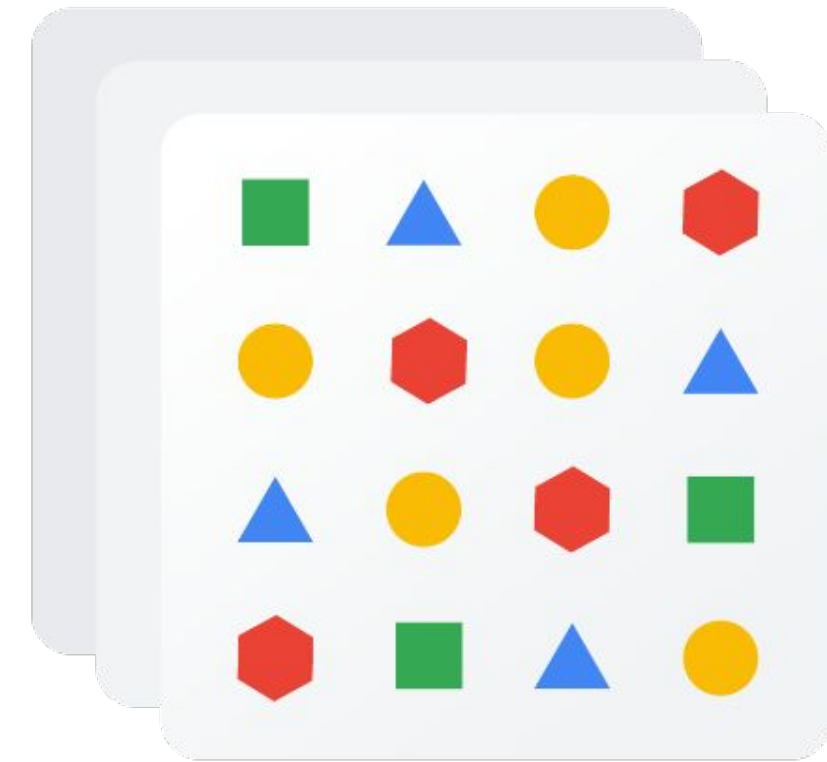
BigQuery ML brings machine learning to the data



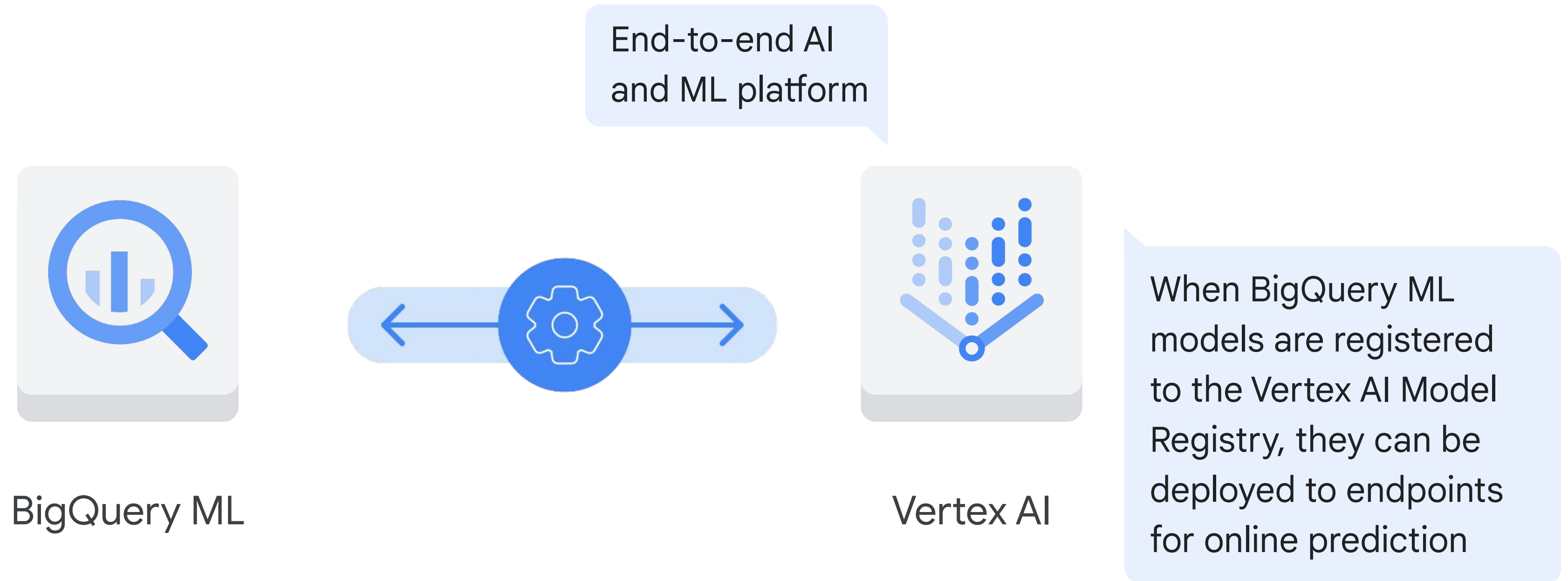
BigQuery ML



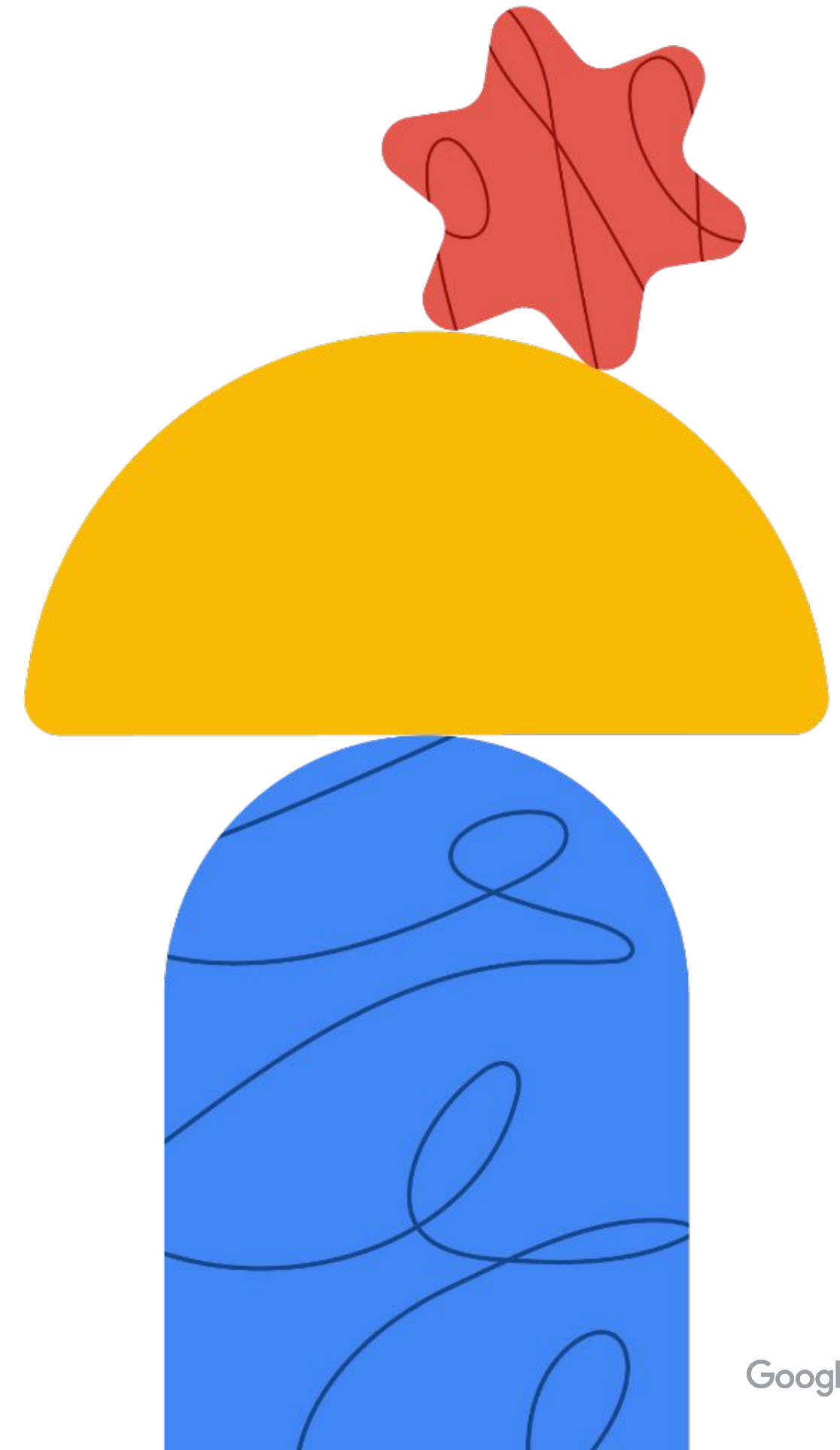
- ✓ Requires fewer tools.
- ✓ Increases speed to production.



BigQuery ML integrates with Vertex AI



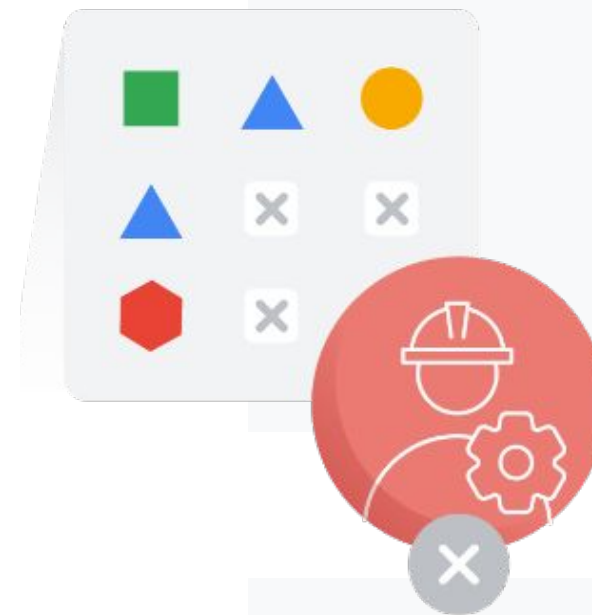
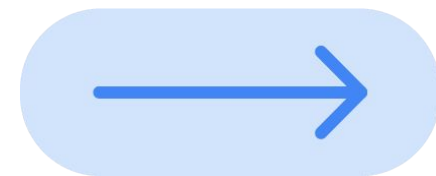
Pre-trained APIs



Pre-trained APIs are a great option if you don't have your own training data



Pre-trained APIs



Ideal in situations where an organization **doesn't** have specialized data scientists...

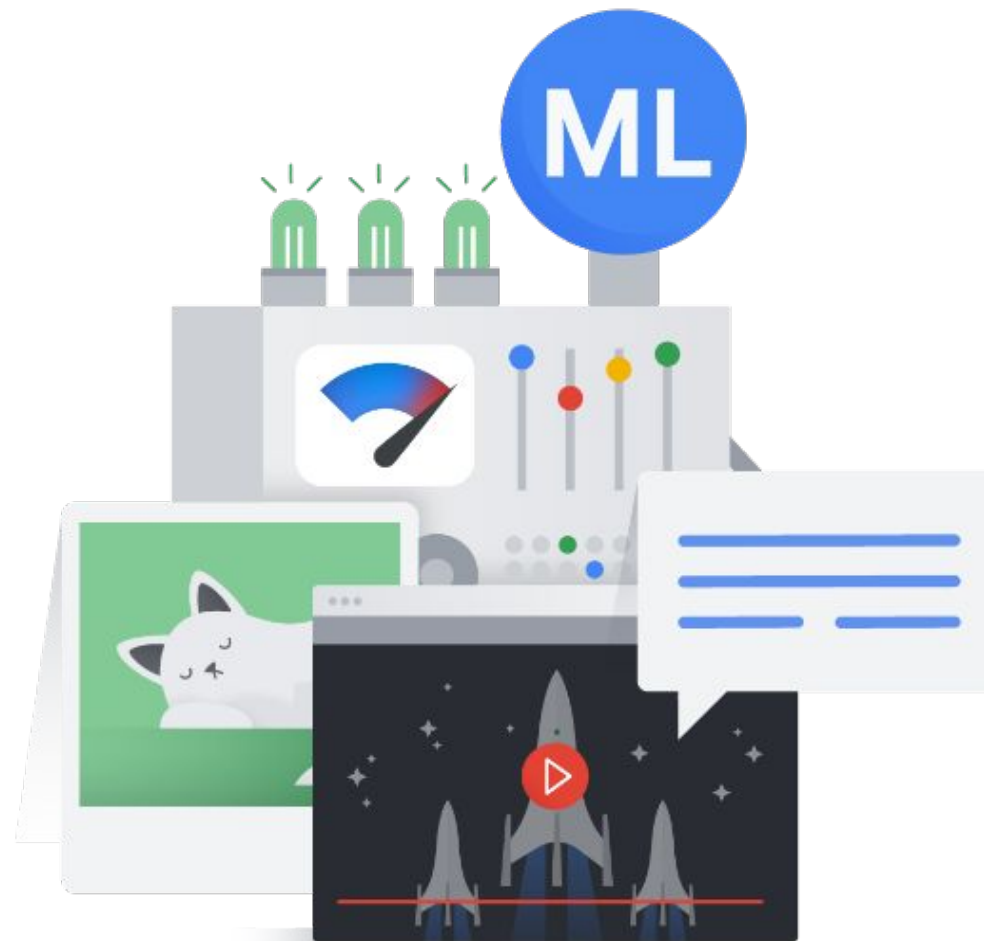
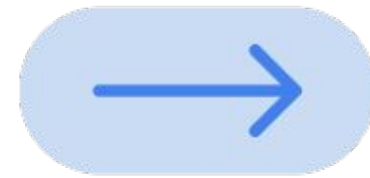


...but it **does** have business analysts and developers

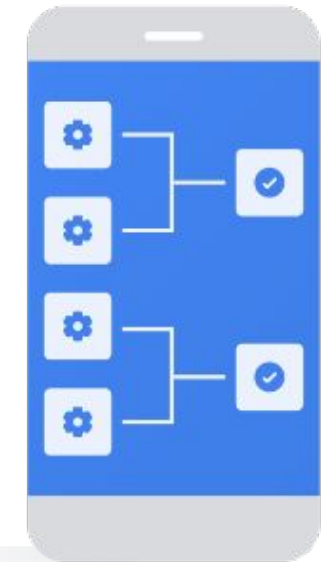
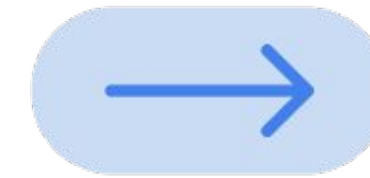
Pre-trained APIs can help developers build smart apps quickly



Pre-trained APIs



Google Cloud provides
access to ML models for
common tasks



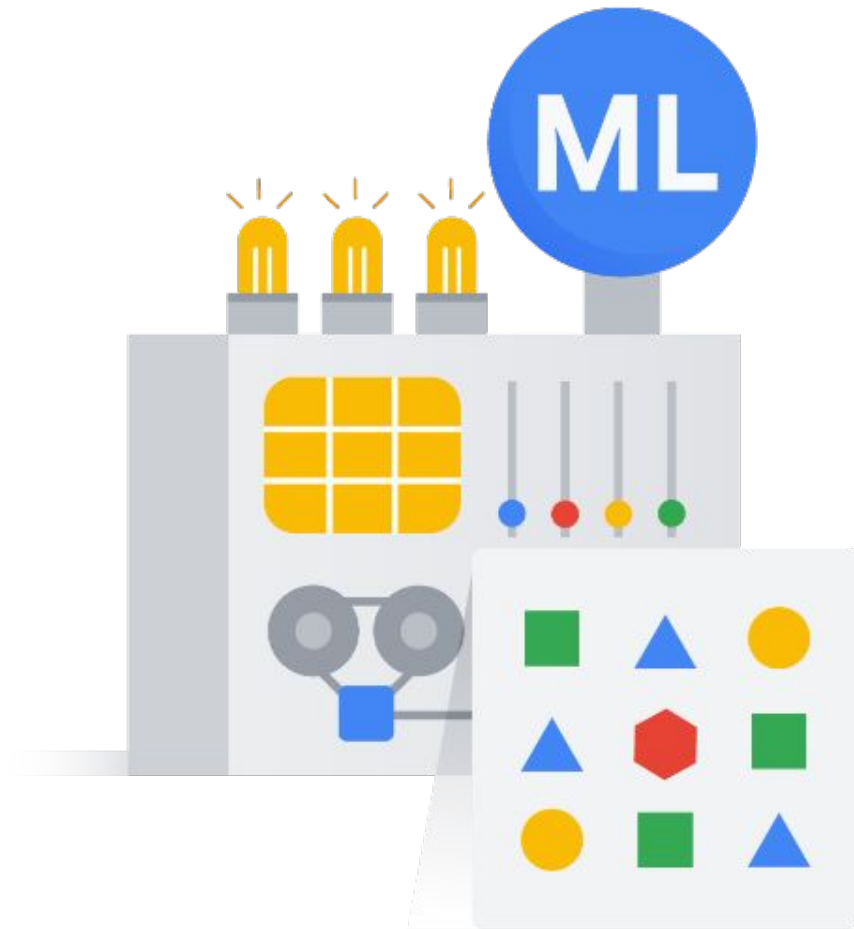
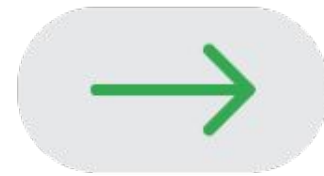
Build apps

Example: A mobile app that users submit photos to

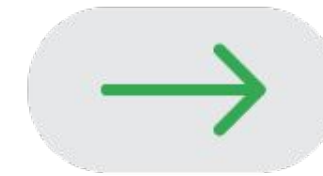
App needs to recognize what the images are and filter out any that aren't safe for work



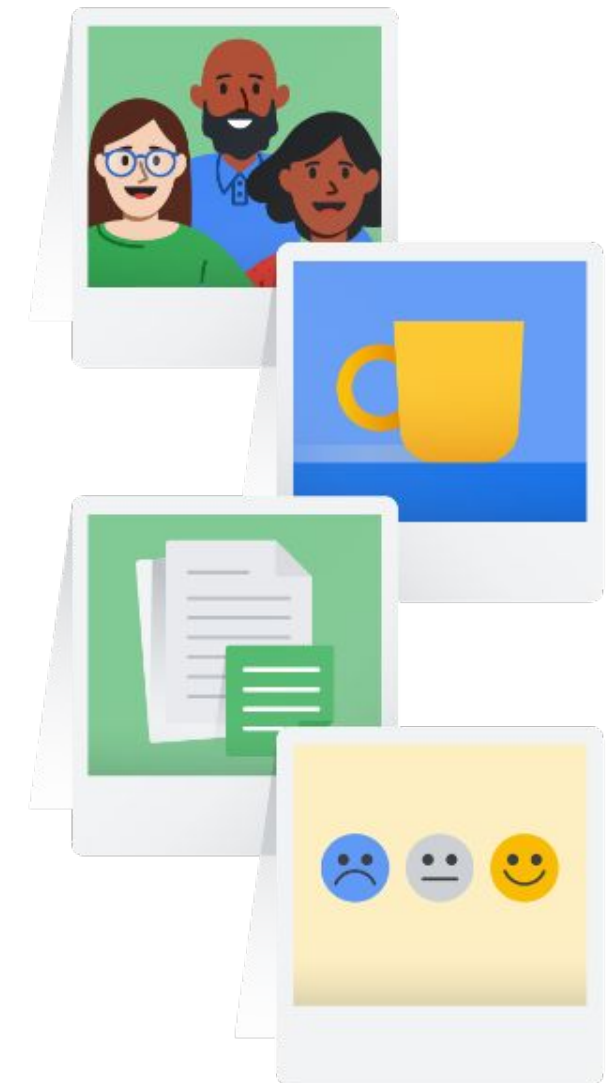
Vision API



Uses pre-trained ML models, which use Google data



Assigns labels



Natural Language API

The API can be used to:

- ✓ Discover syntax, entities, and sentiment in text.
- ✓ Classify text into a predefined set of categories.



A business has a contact form on its website that receives many daily messages

Google's pre-trained APIs



The **Cloud Translation API** converts text from one language to another.



The **Speech-to-Text API** converts audio to text for data processing.

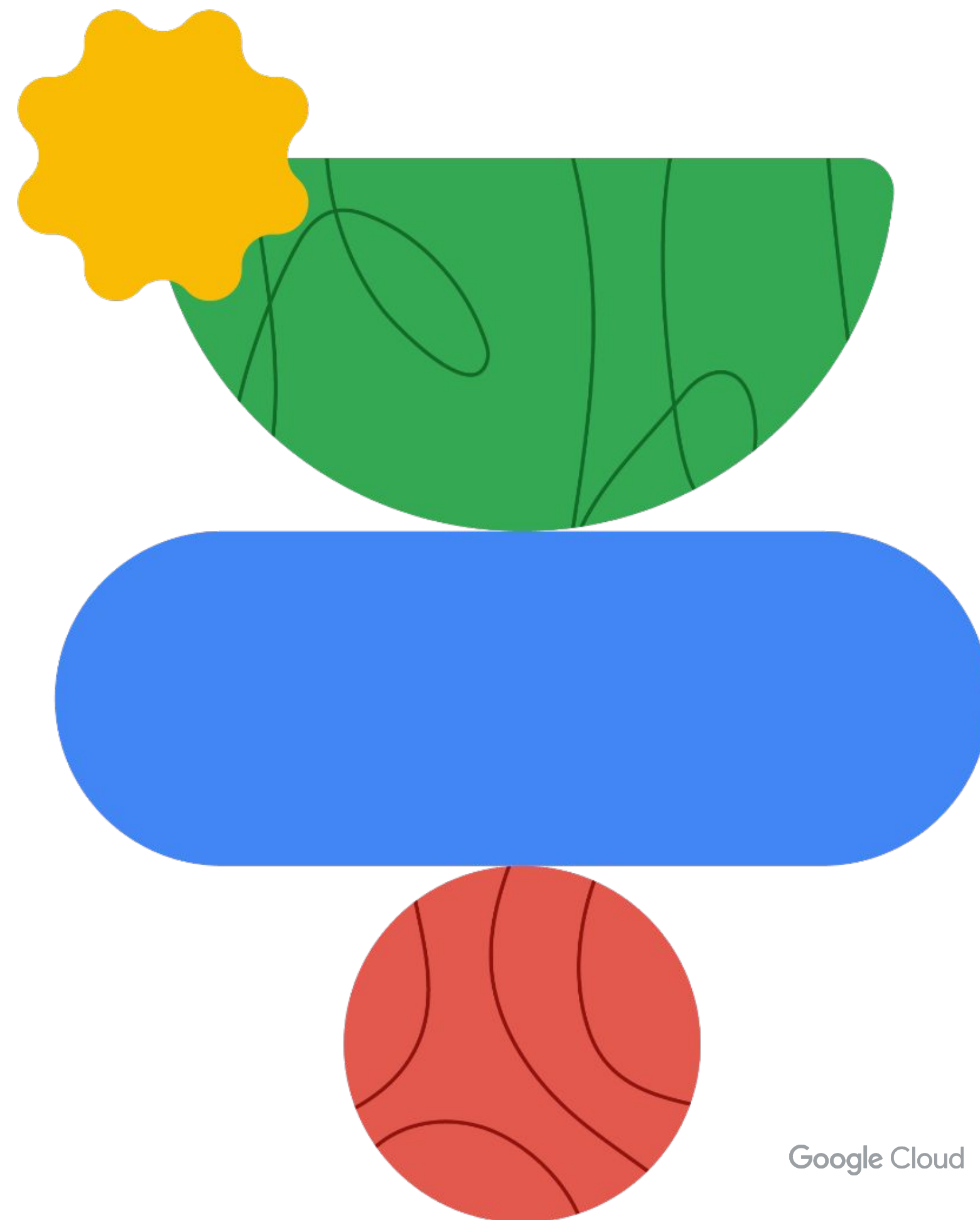


The **Text-to-Speech API** converts text into high-quality voice audio.



The **Video Intelligence API** recognizes motion and action in video.

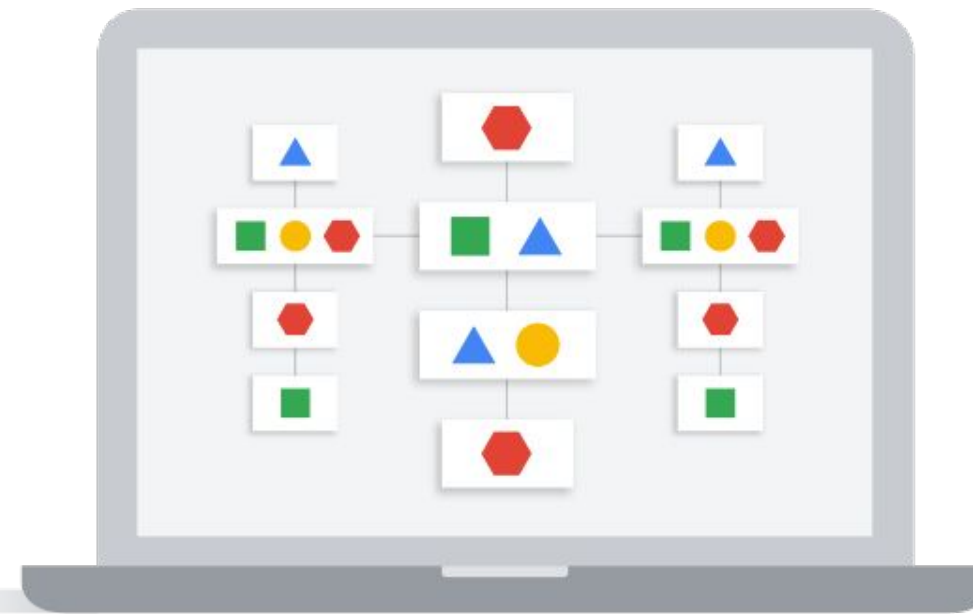
AutoML



Training models with your own data



Your data



Vertex AI



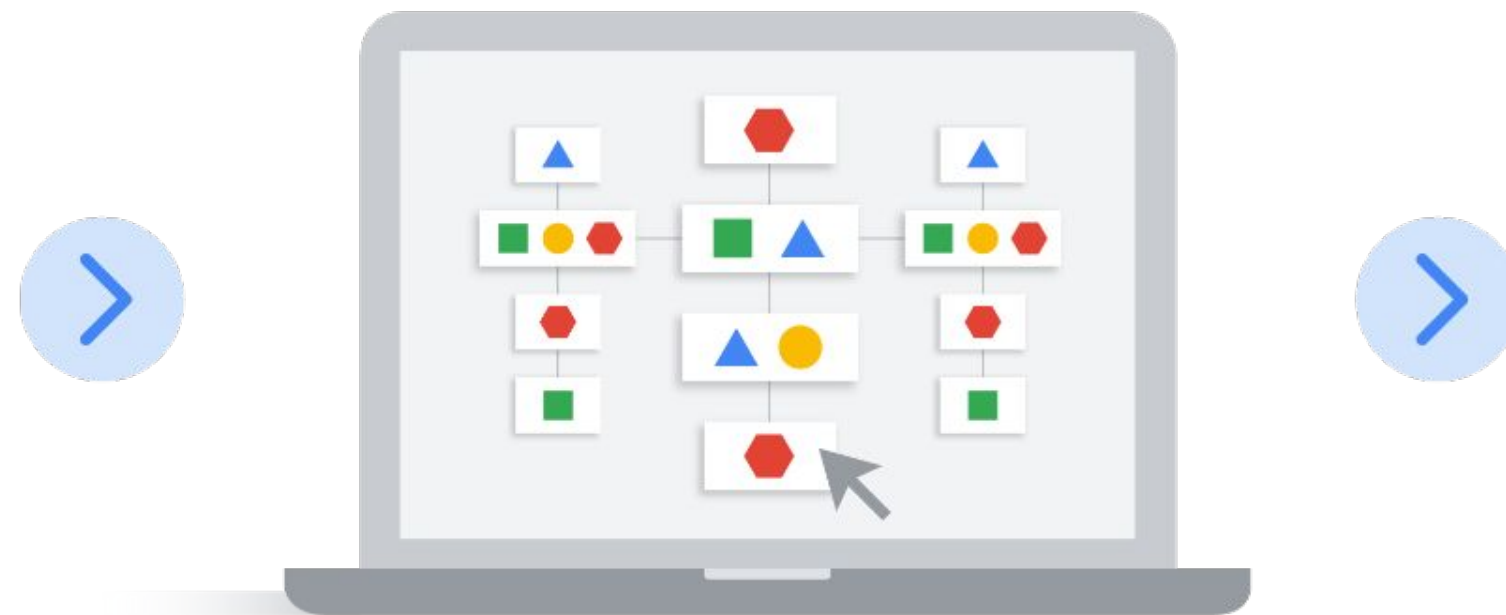
ML projects

AutoML on Vertex AI lets you build and train ML models from end-to-end using GUIs

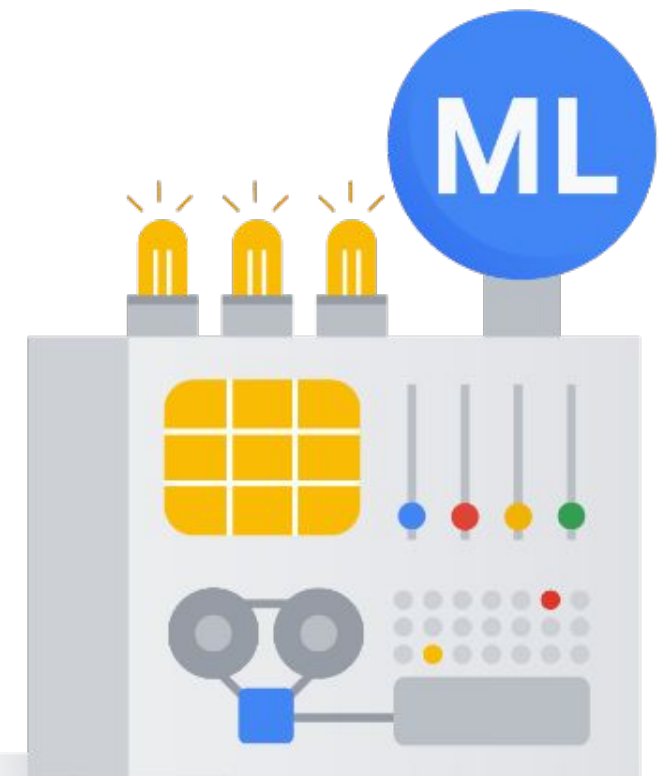
AutoML



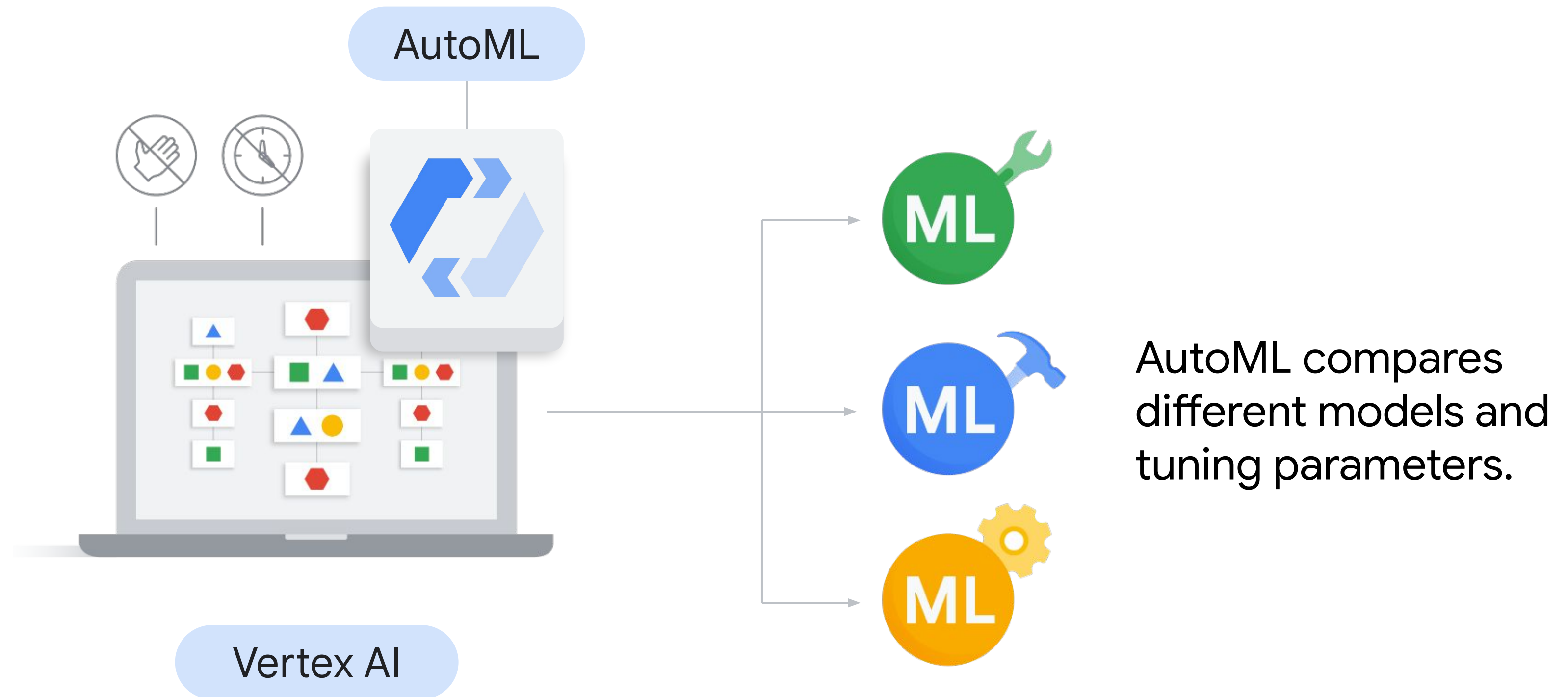
Graphical user interfaces



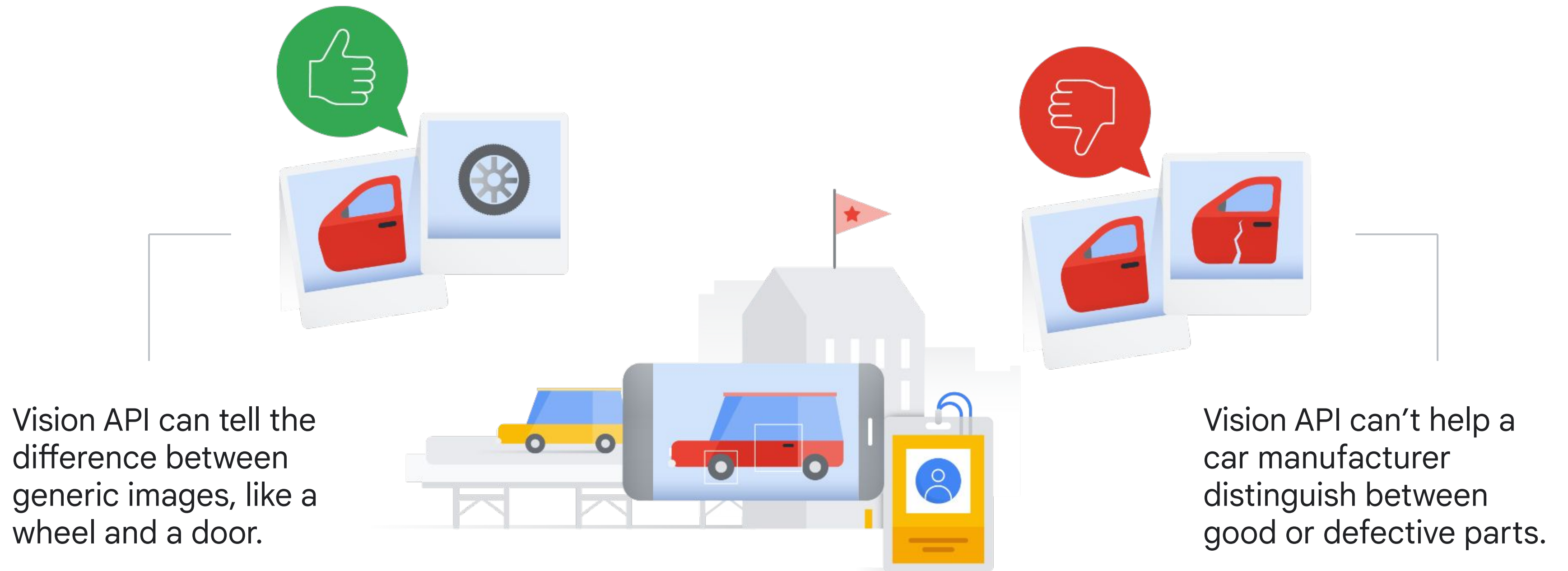
No code



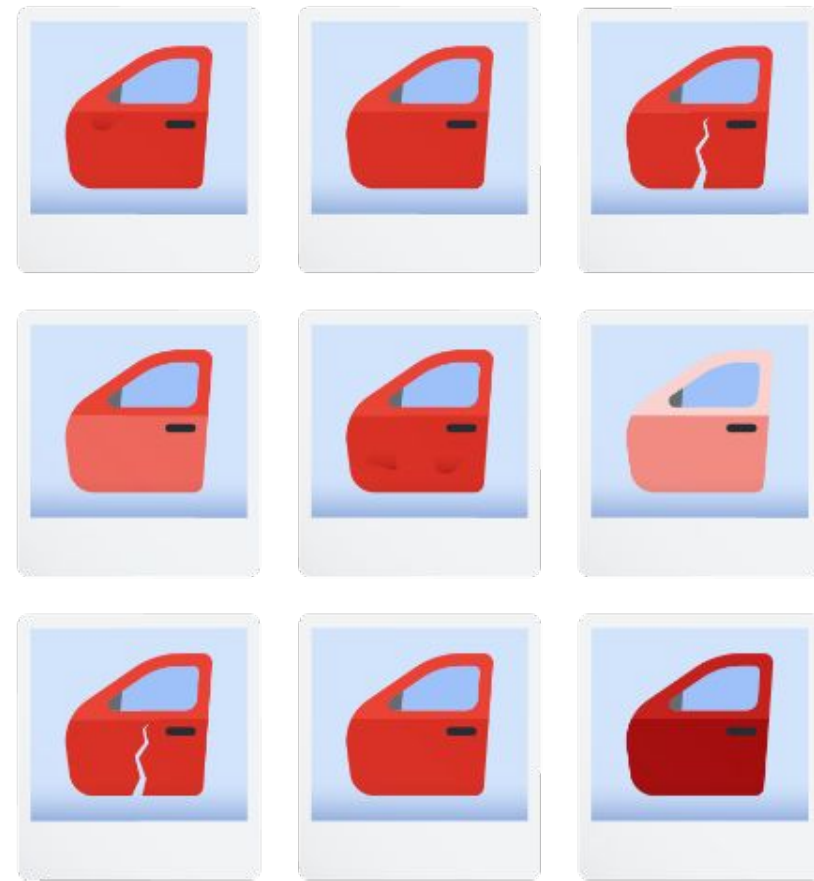
AutoML chooses the best machine learning model for you



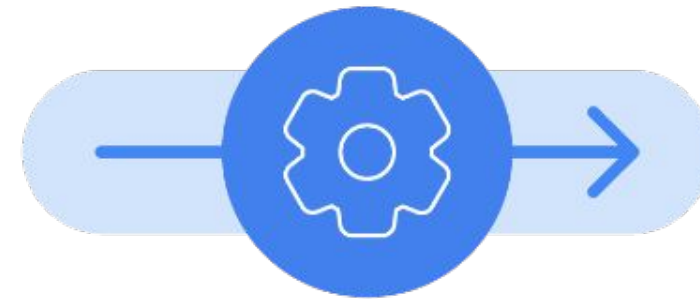
Back to the image recognition example: Vision API



AutoML Vision can be trained on specialized data

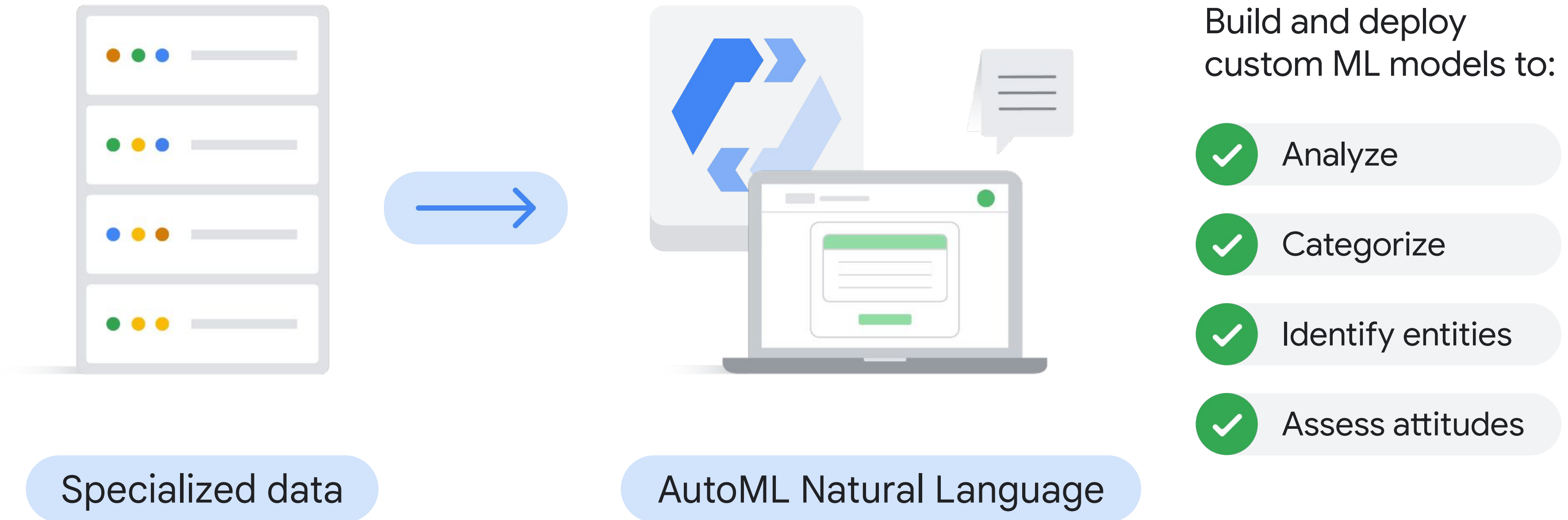


Upload a batch of image
and train a classification model
through the AutoML GUI

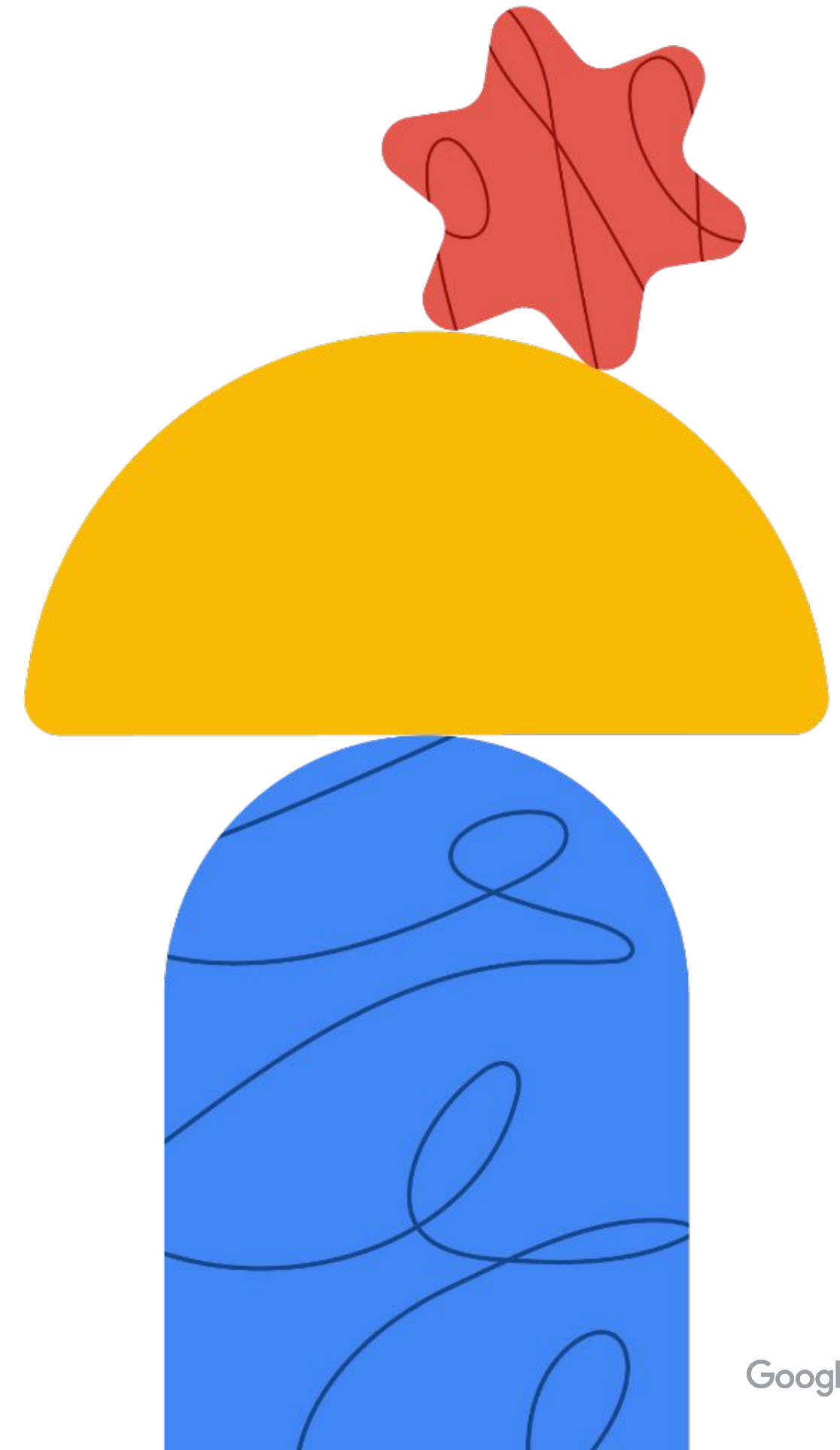


AutoML Vision

AutoML feature: AutoML Natural Language



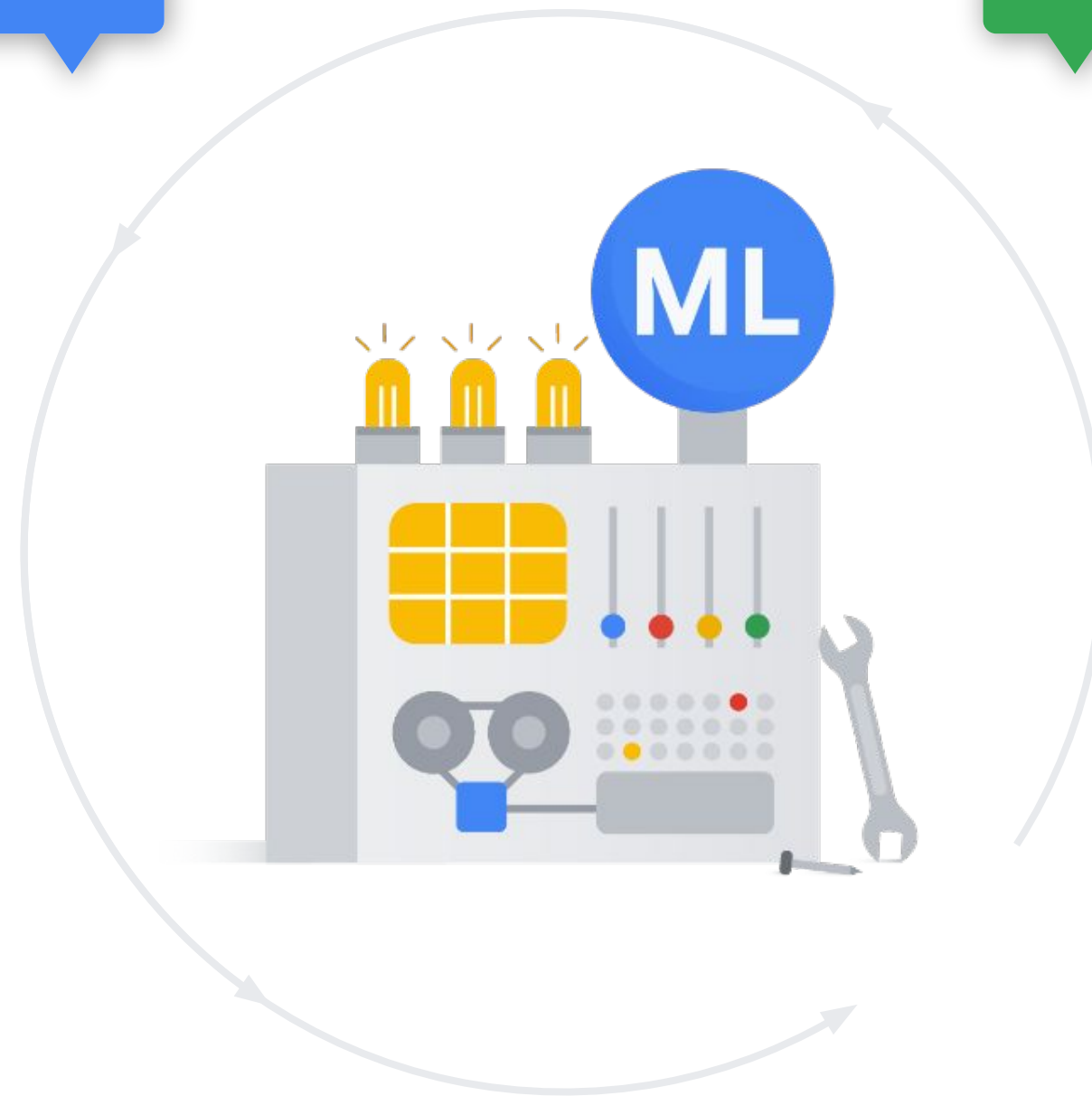
Custom models



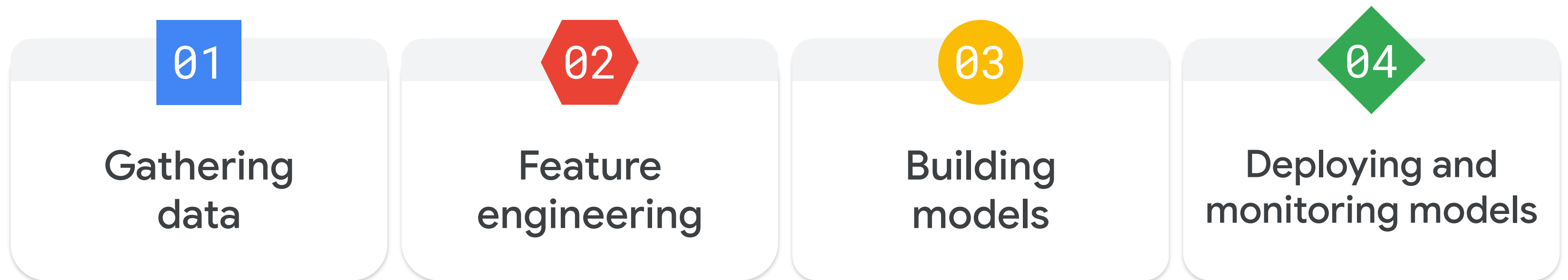
Custom models can be built with Vertex AI

Trained with your own data

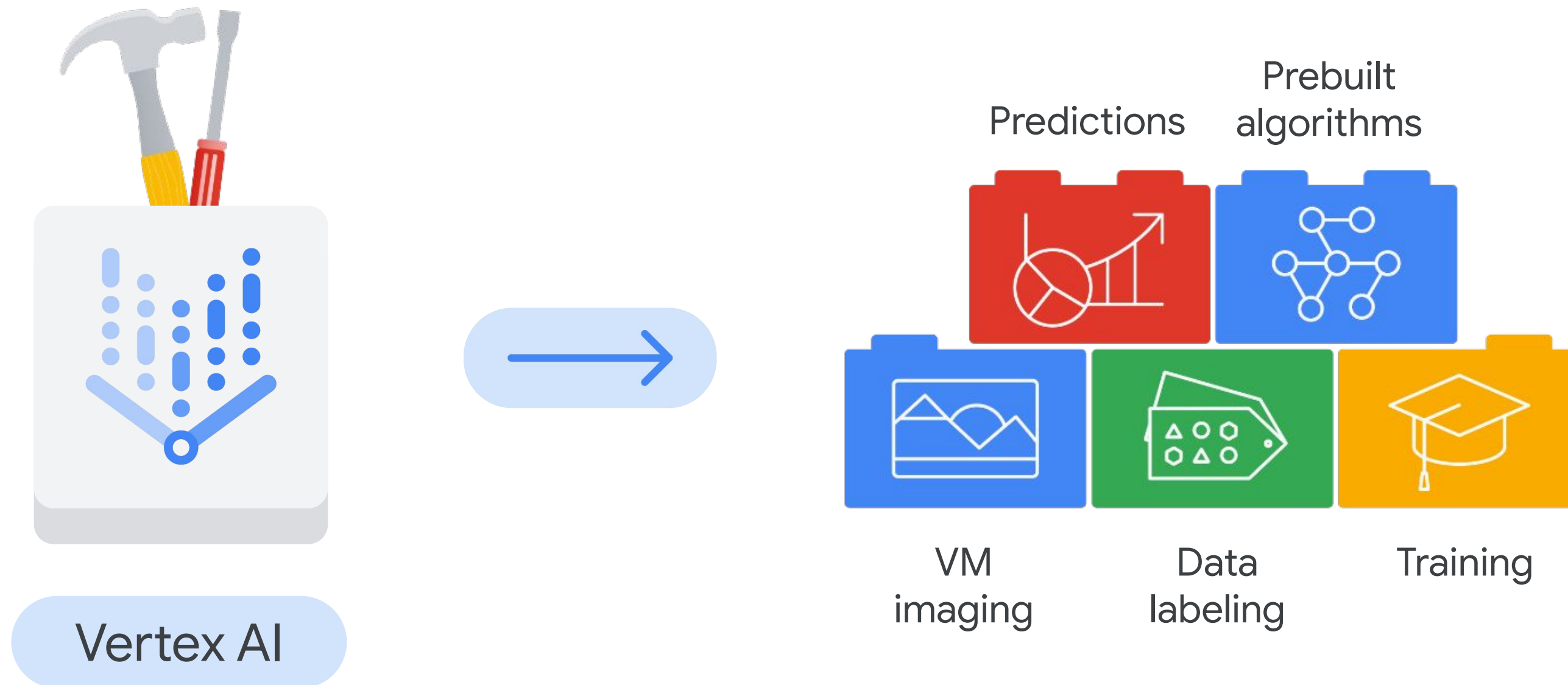
Custom-built



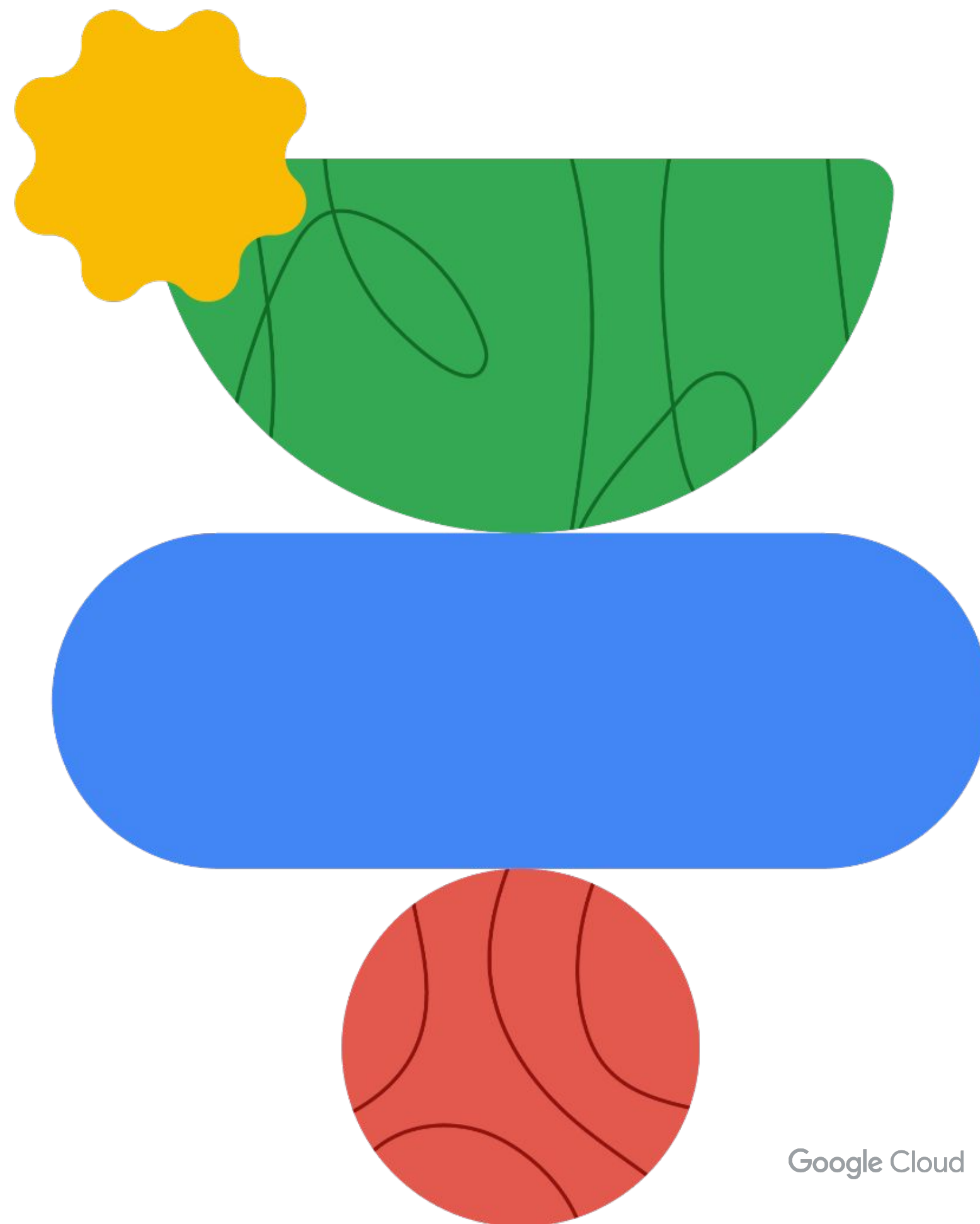
Vertex AI provides a suite of products to help at each stage of the ML workflow



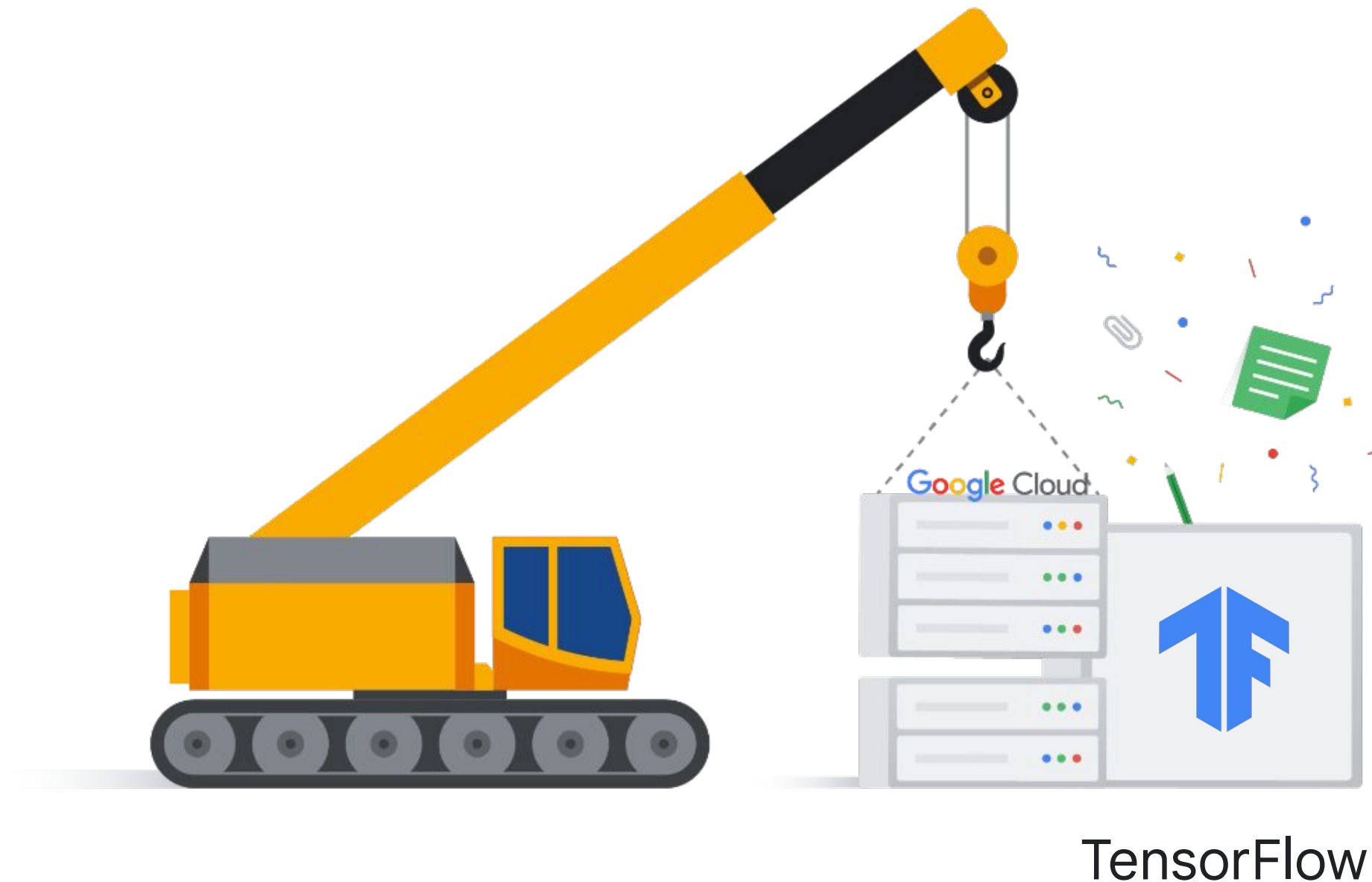
Vertex AI contains tools that assist programmers



TensorFlow



An end-to-end open source ML platform

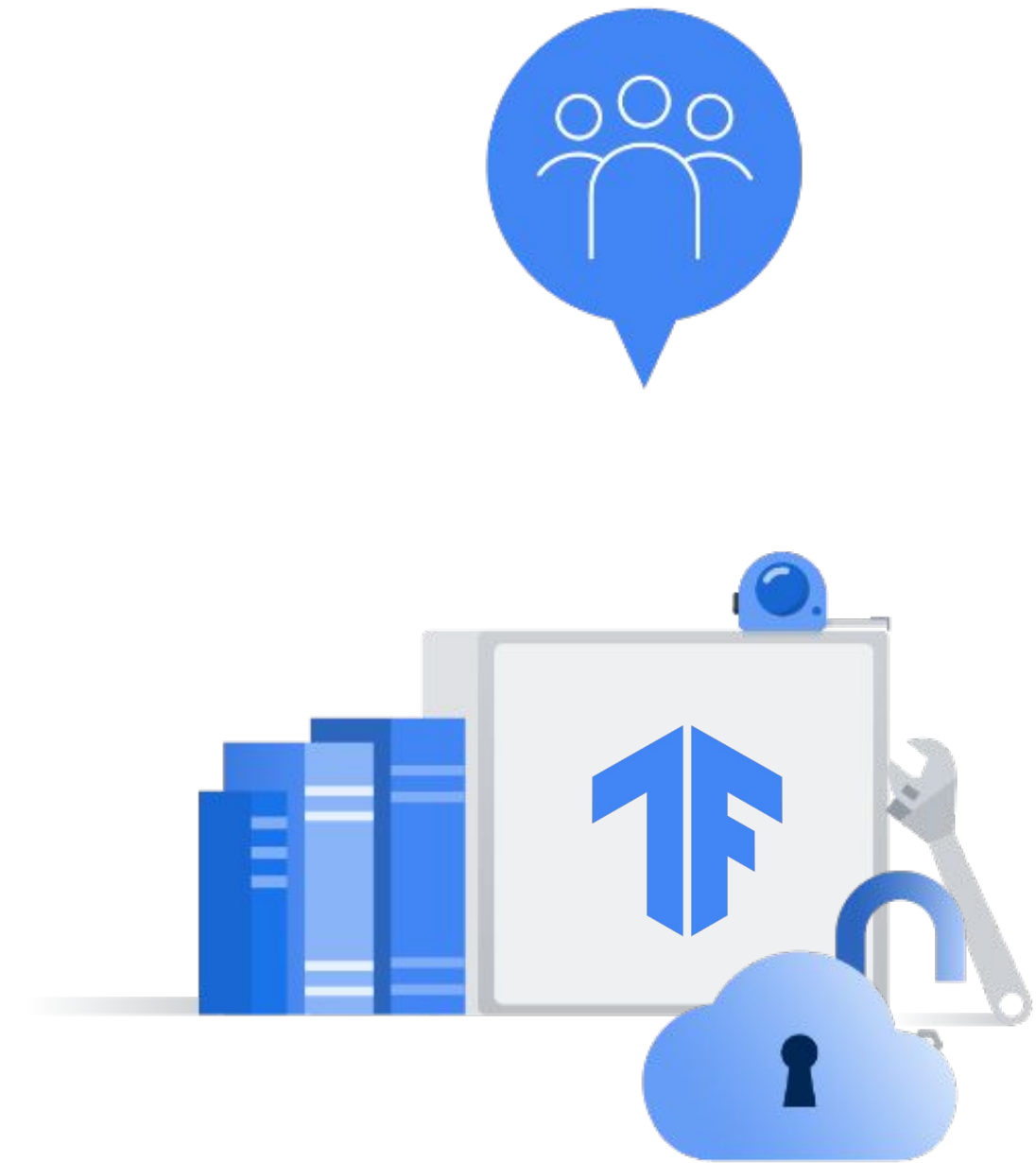


All machine learning models are built on top of Google Cloud's AI foundational infrastructure.

A part of this foundation is TensorFlow.

TensorFlow has a flexible ecosystem of tools, libraries, and community resources

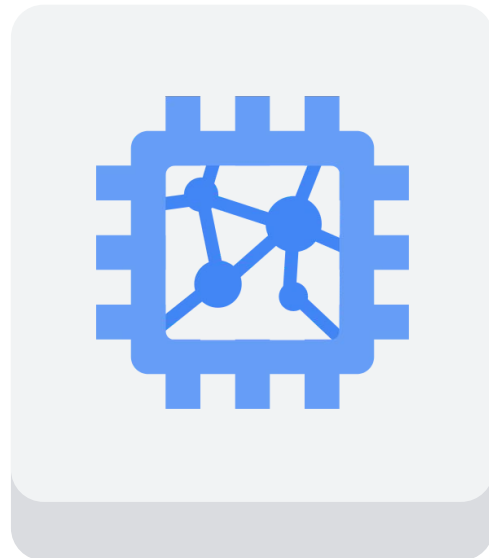
- ✓ Allows researchers to innovate in ML
- ✓ Allows developers to build and deploy ML powered applications



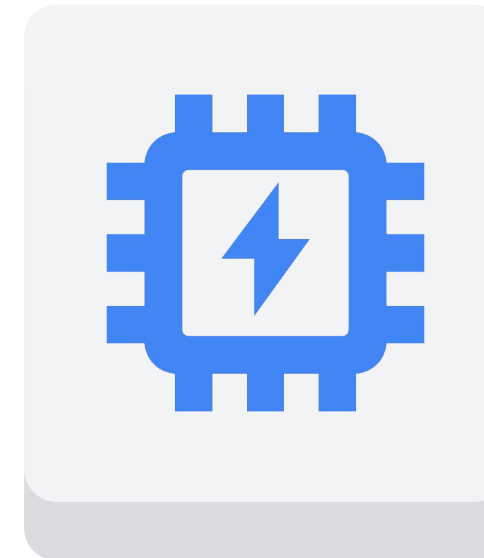
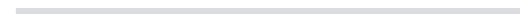
TensorFlow uses the Tensor Processing Unit (TPU)

ASIC

Google's custom-developed application-specific integrated circuit

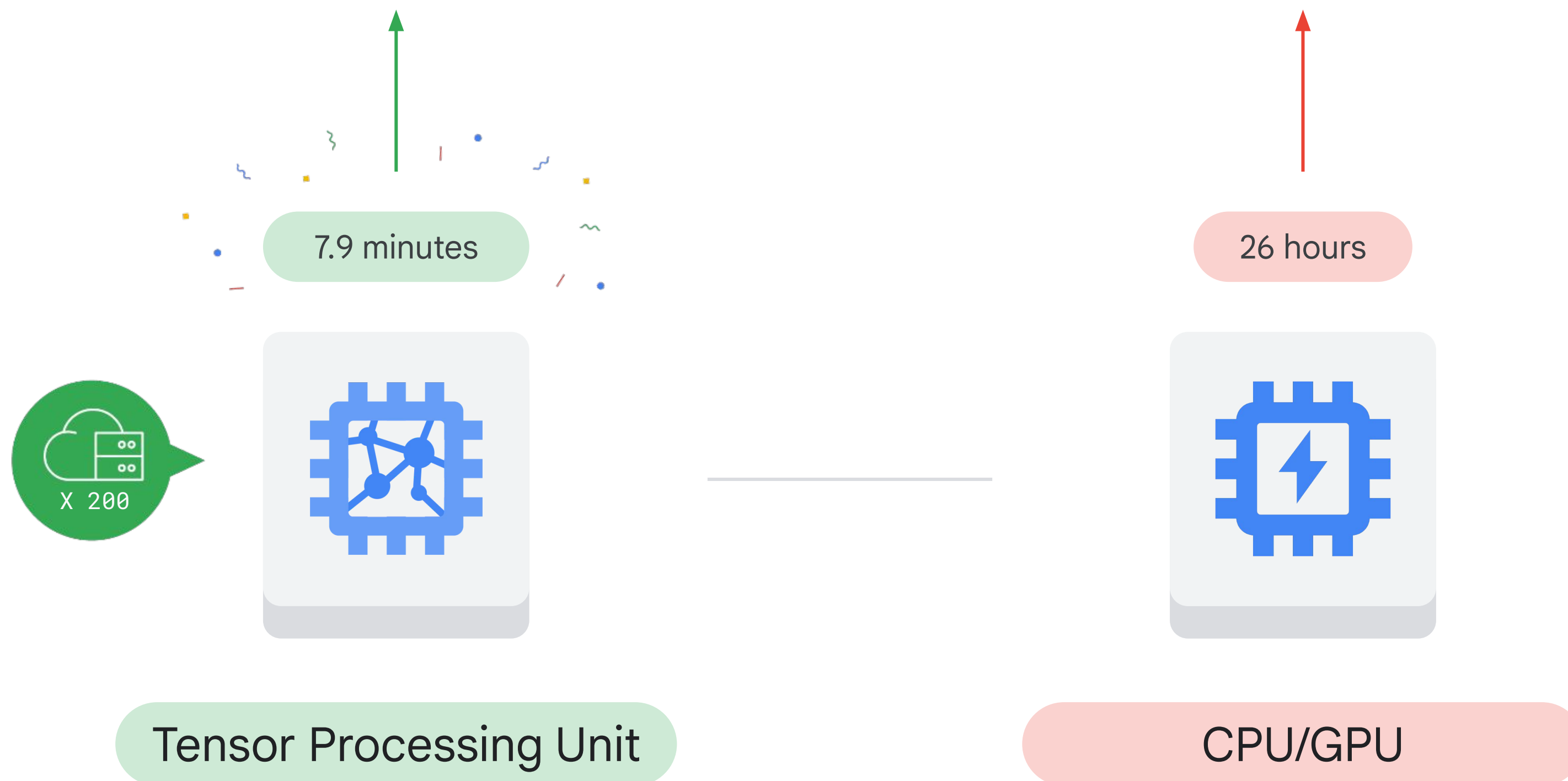


Tensor Processing Unit



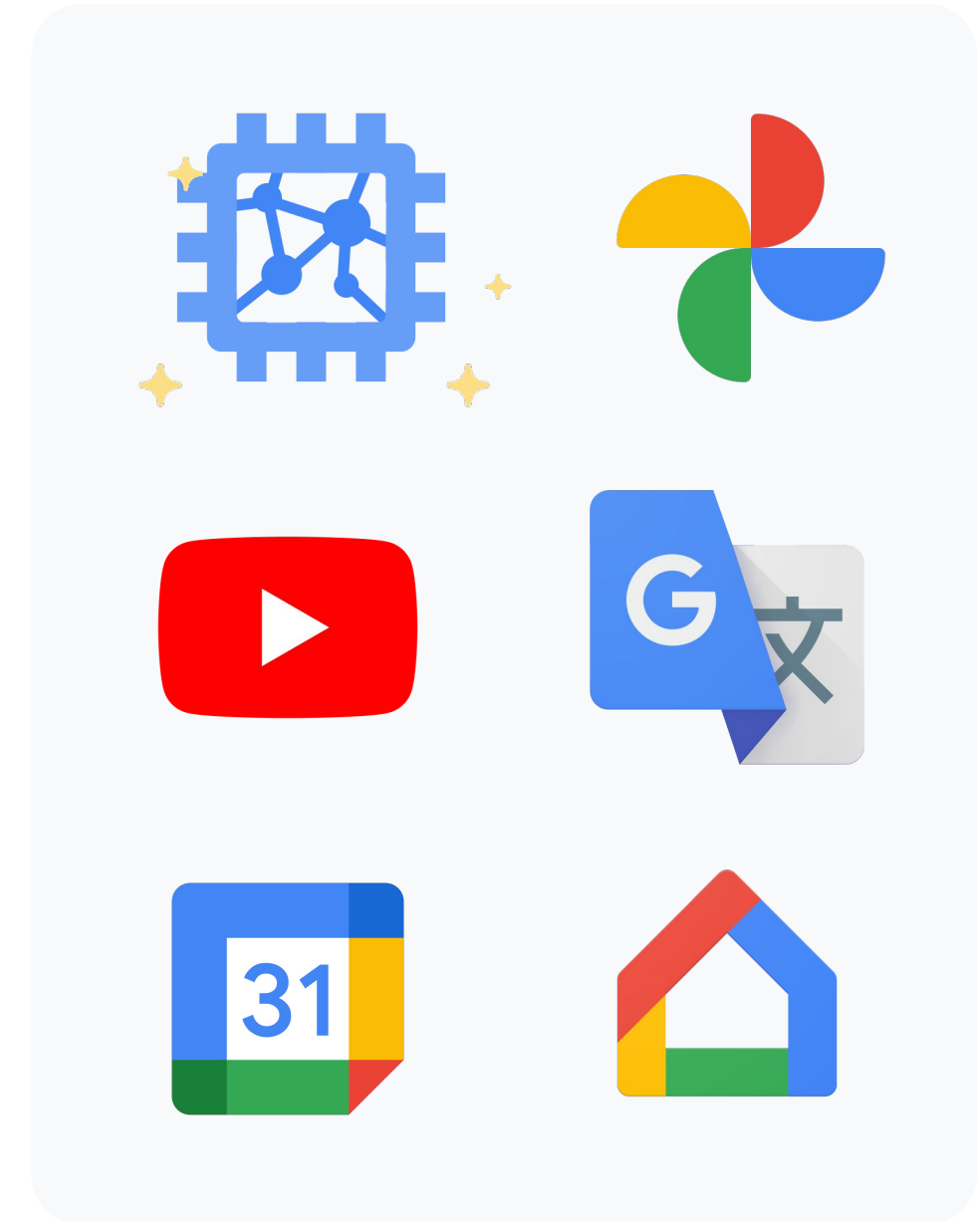
CPU/GPU

Computing speed is fast with a TPU

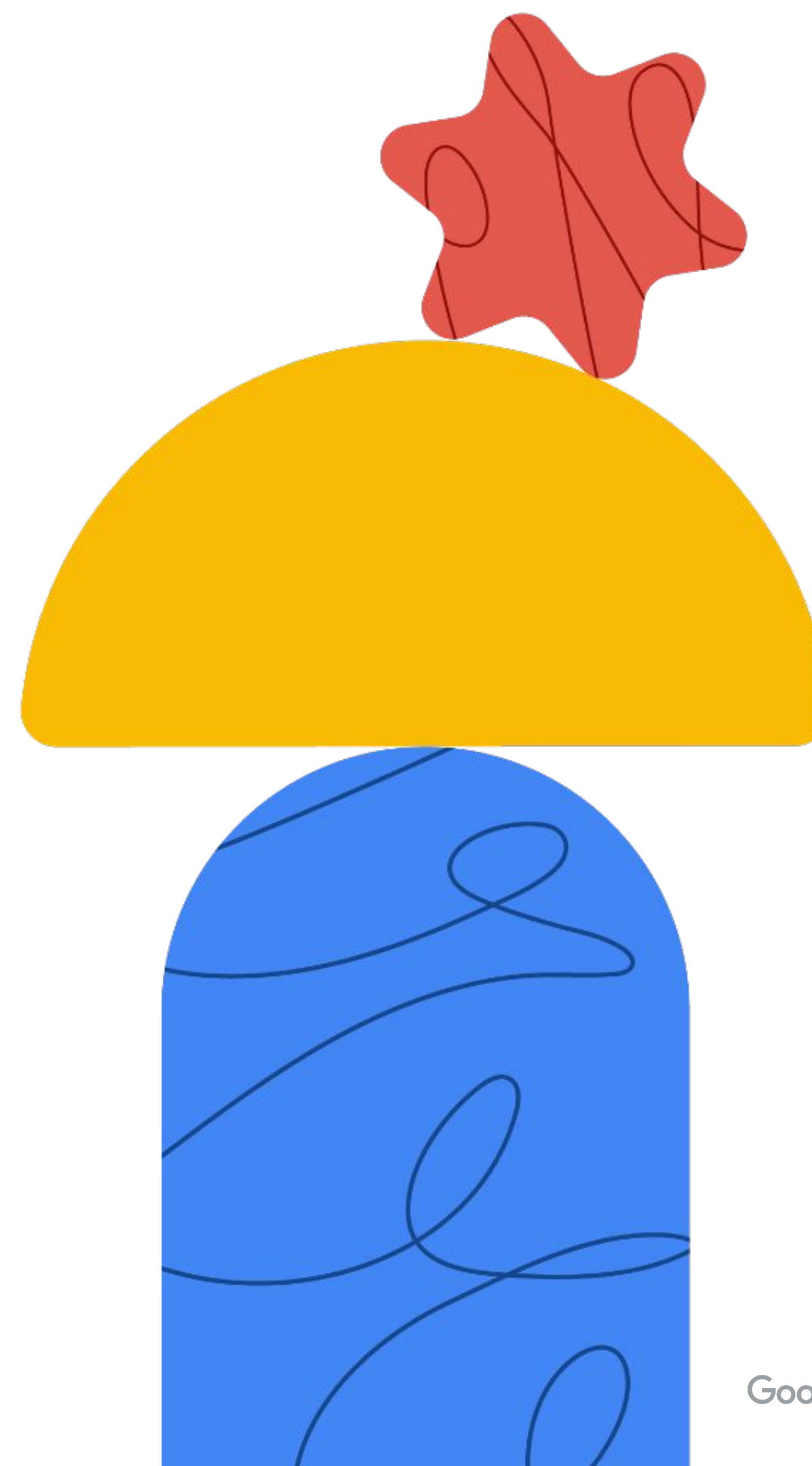


Cloud TPUs have been integrated across Google products

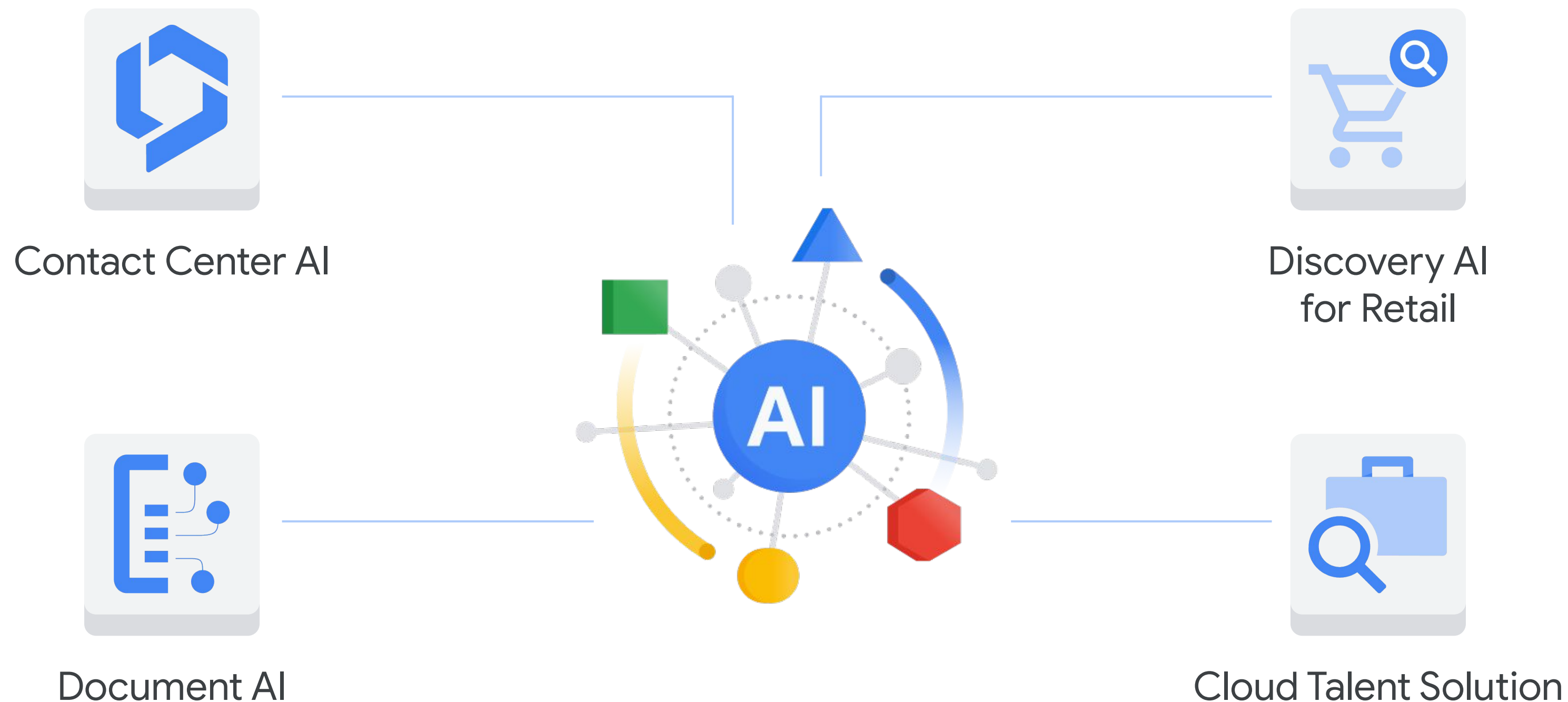
Google Cloud



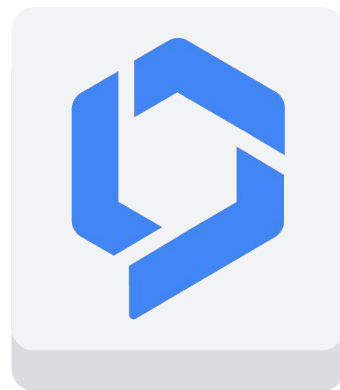
AI solutions



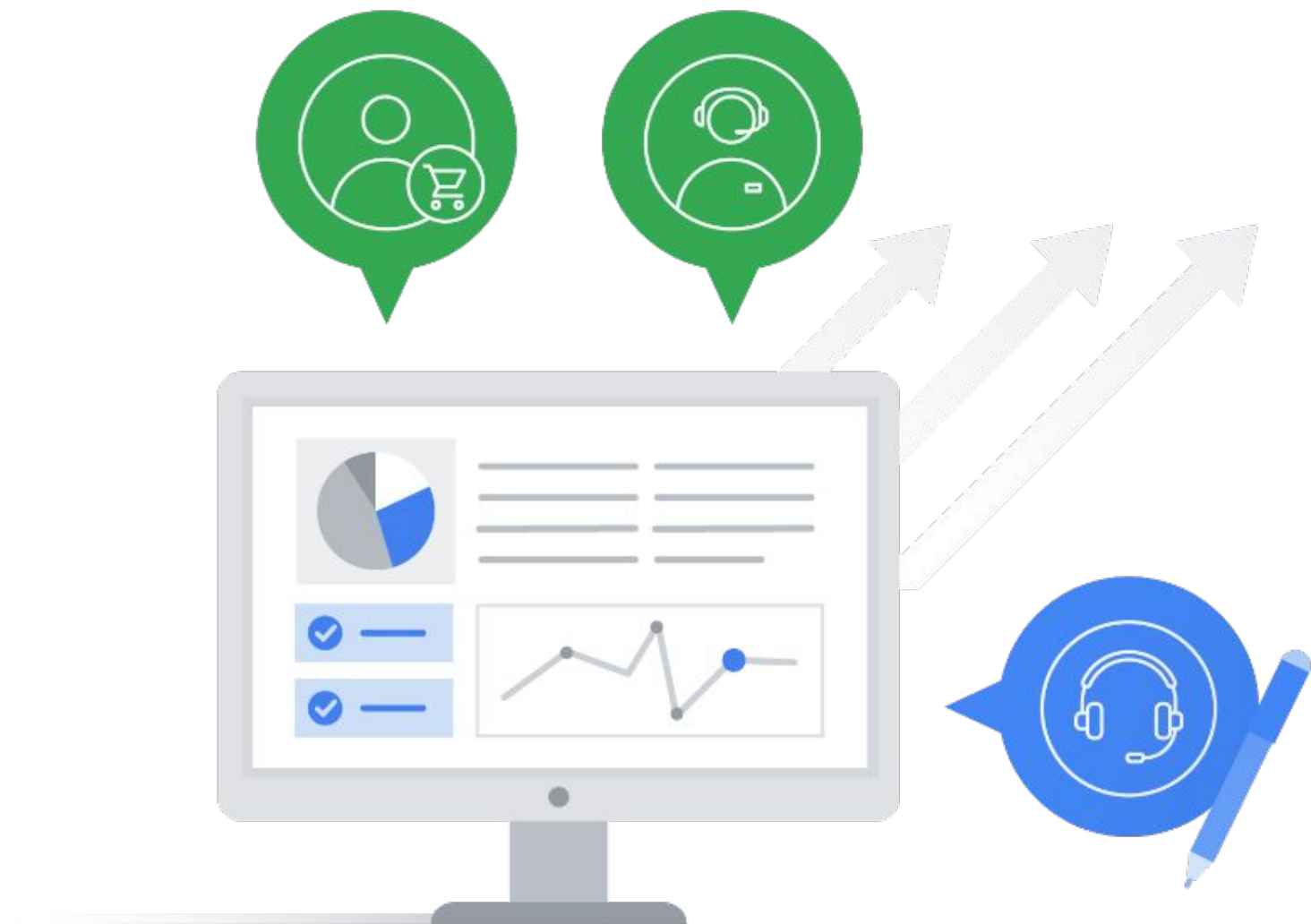
Google Cloud's AI solutions



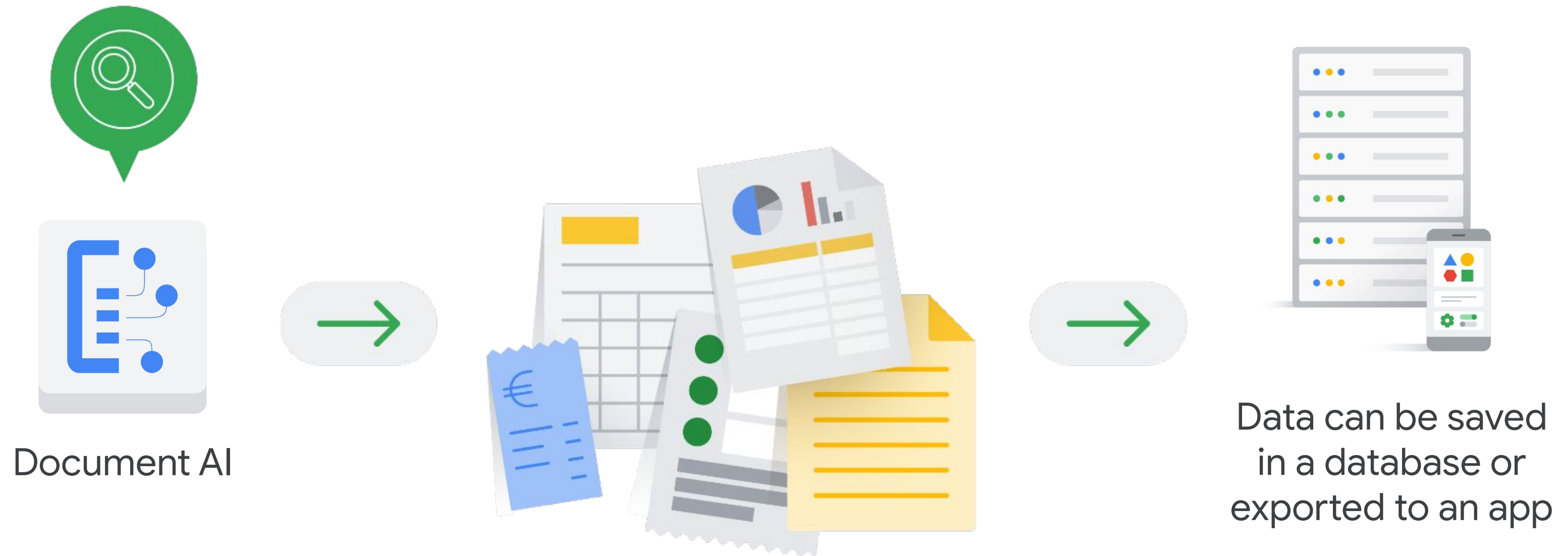
Contact Center AI provides models for speaking with customers and assisting human agents



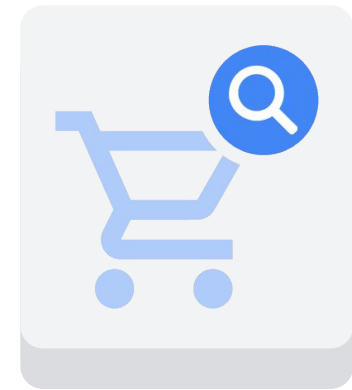
Contact Center AI



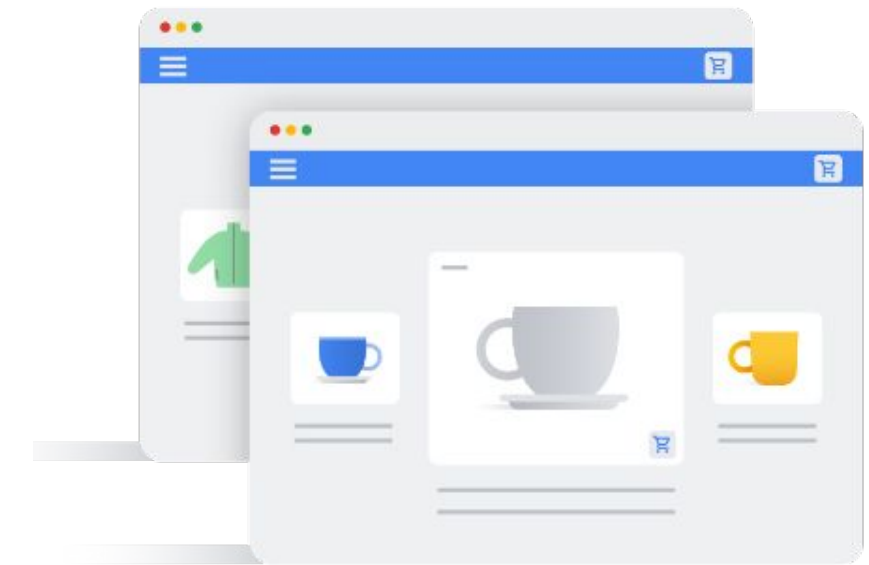
Document AI extracts and classifies information from unstructured documents



Discovery AI for Retail uses ML to order products on a retailer's ecommerce site

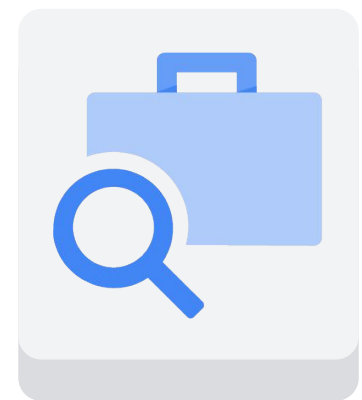


Discovery AI
for Retail



AI learns the ideal
product ordering for
each page on the site.

Cloud Talent Solution uses AI with job search and talent acquisition capabilities

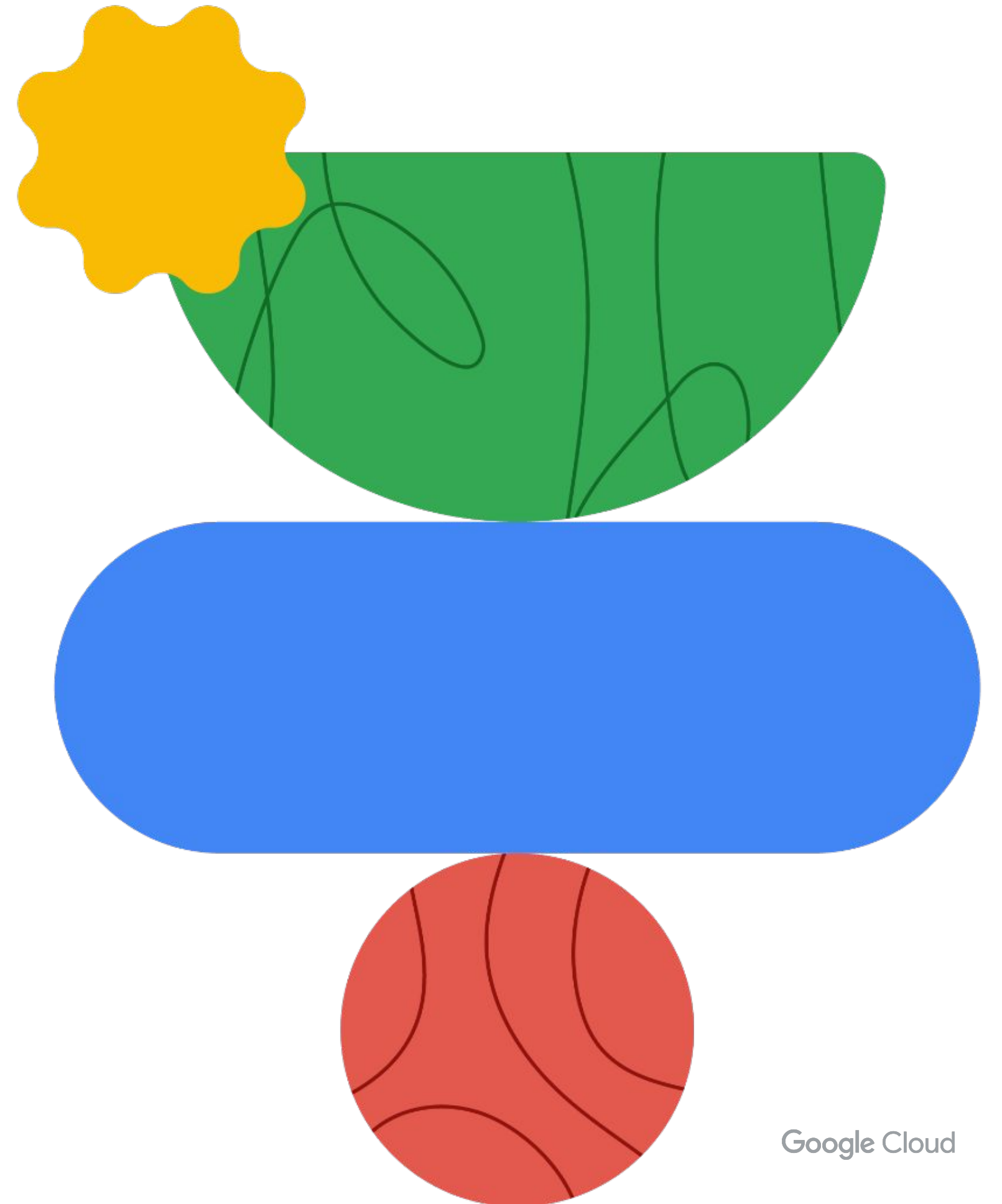


Cloud Talent Solution

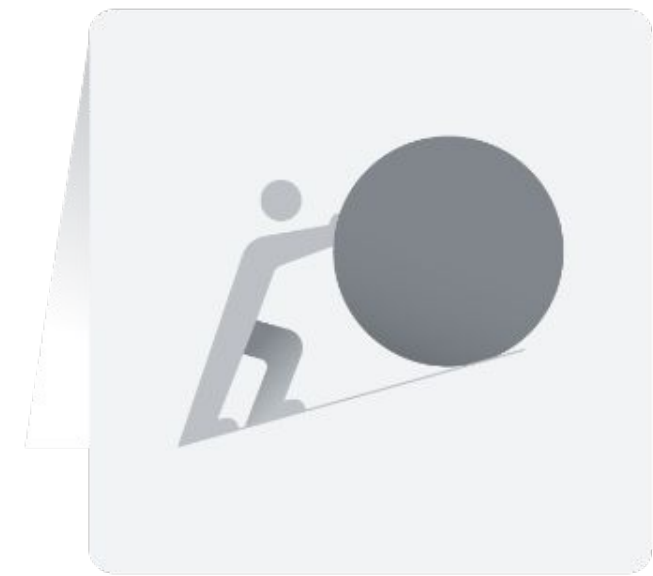
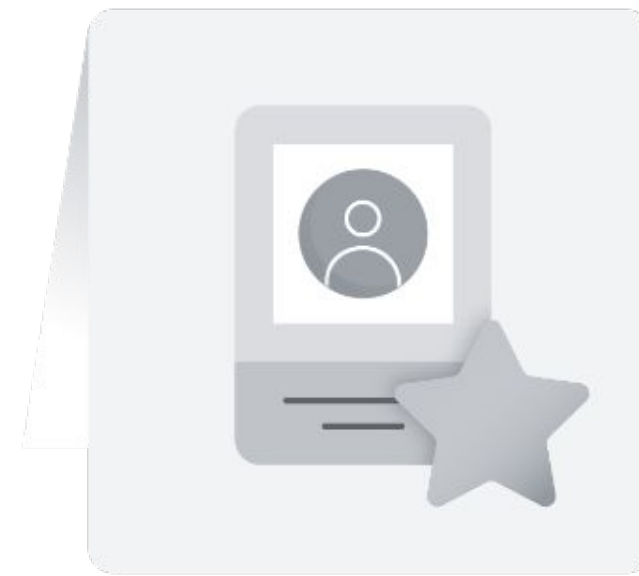
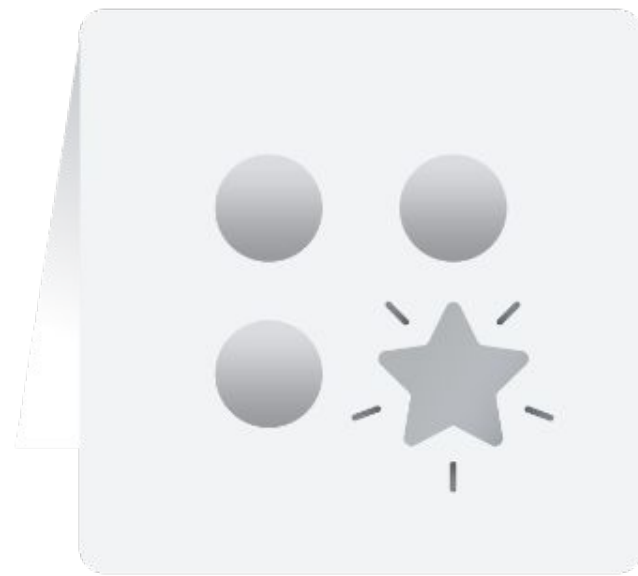


Matches candidates to ideal jobs faster, and allows employers to attract higher quality candidates.

Considerations when selecting Google Cloud AI/ML solutions

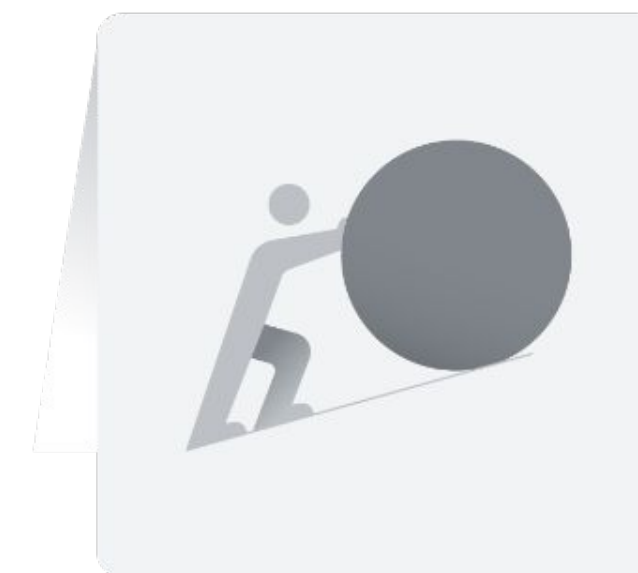
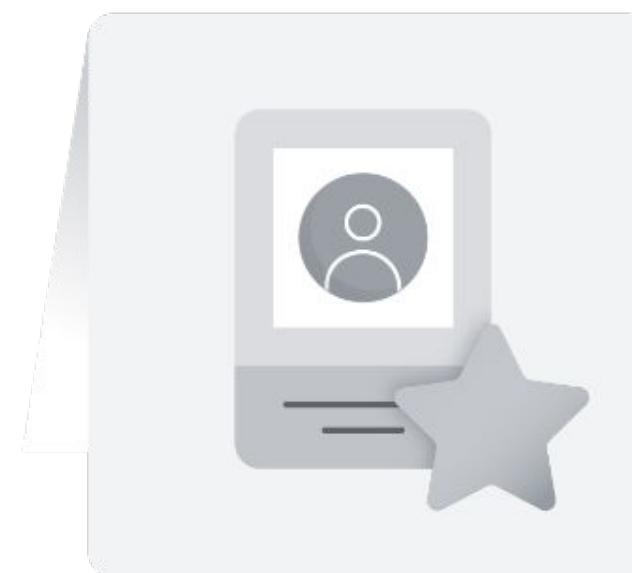
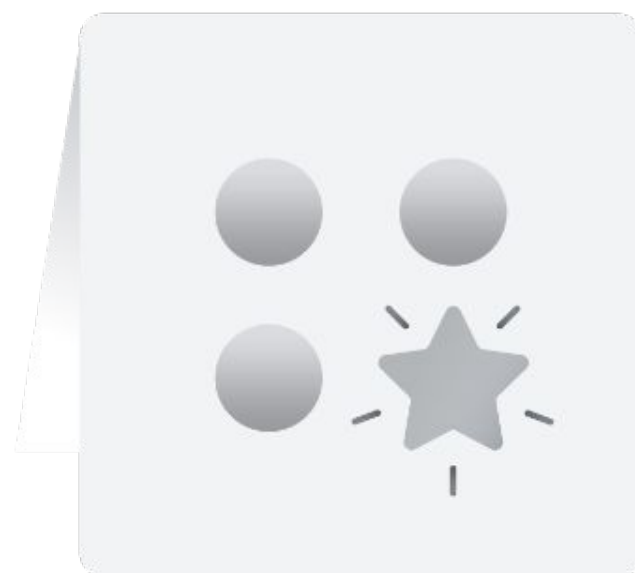
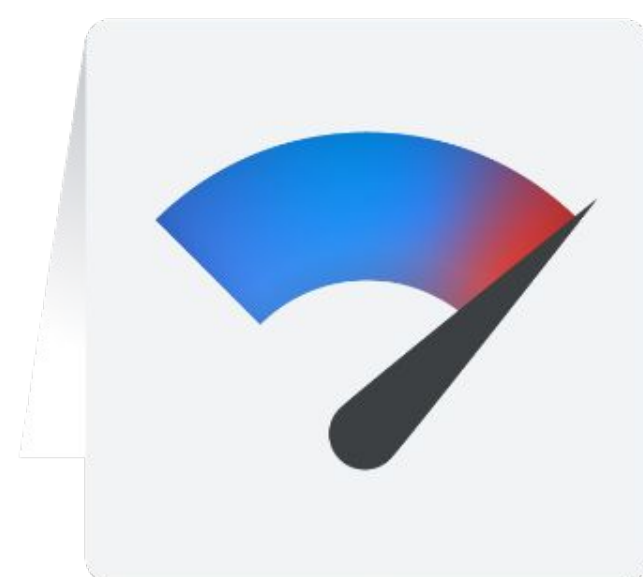


Considerations when selecting which AI or ML solution to employ

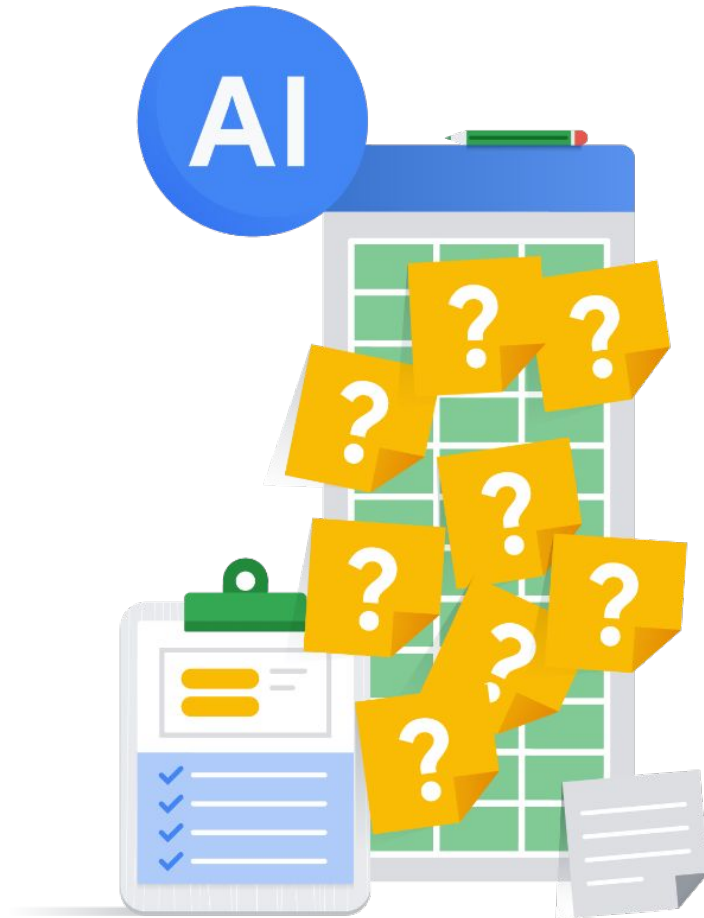


How quickly do you need to get your model to production?

Speed



Speed



AI projects take 3-36 months to plan and implement.



Pre-trained APIs

Quickest: There are no model training requirements.

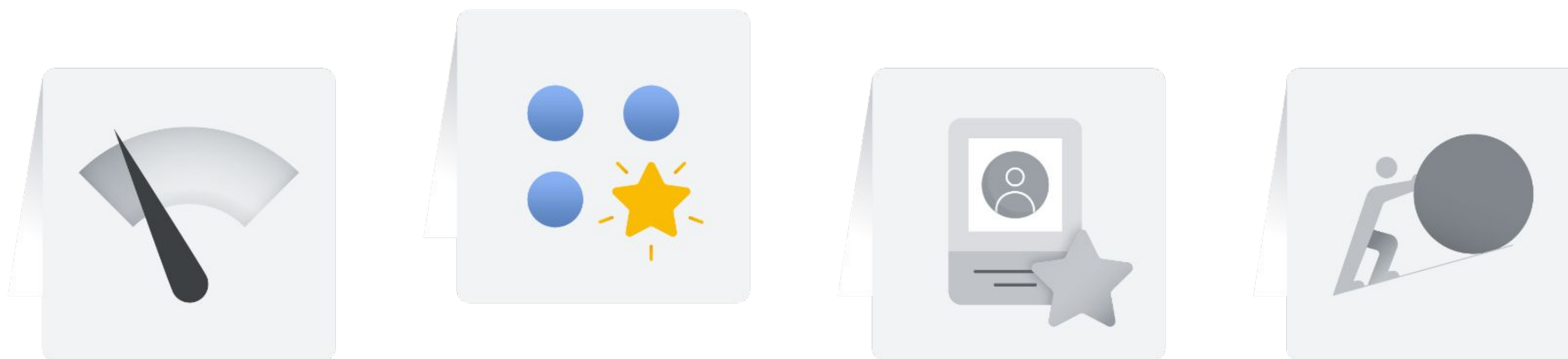


Custom training

Longest: It builds the ML model from the beginning.

How unique does your model need to be?

Differentiation



Differentiation



Google Cloud offers a range of out-of-the-box solutions



They're quick to start.

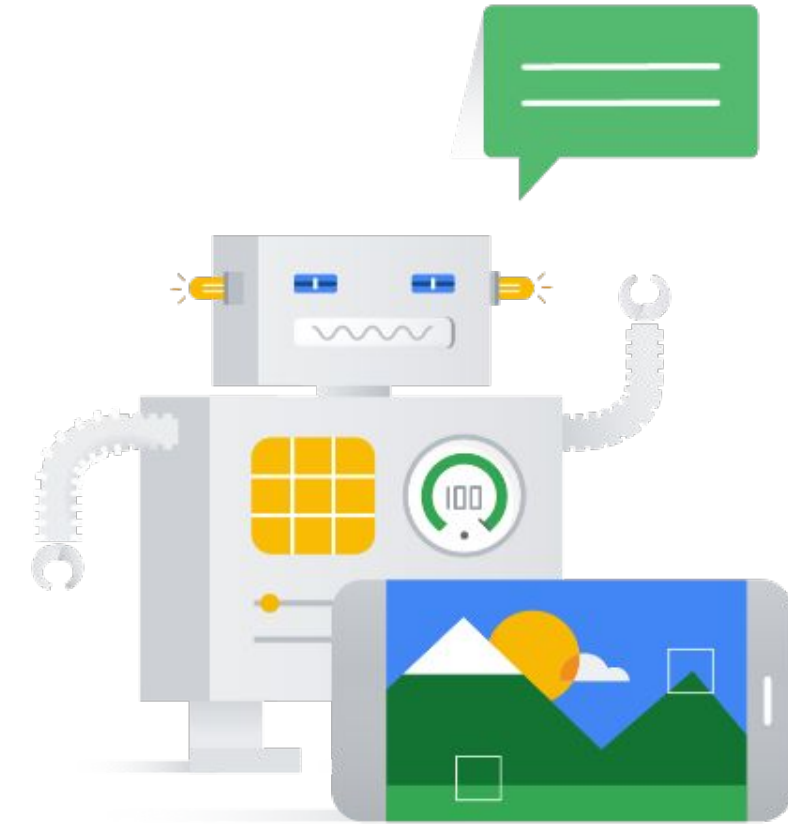
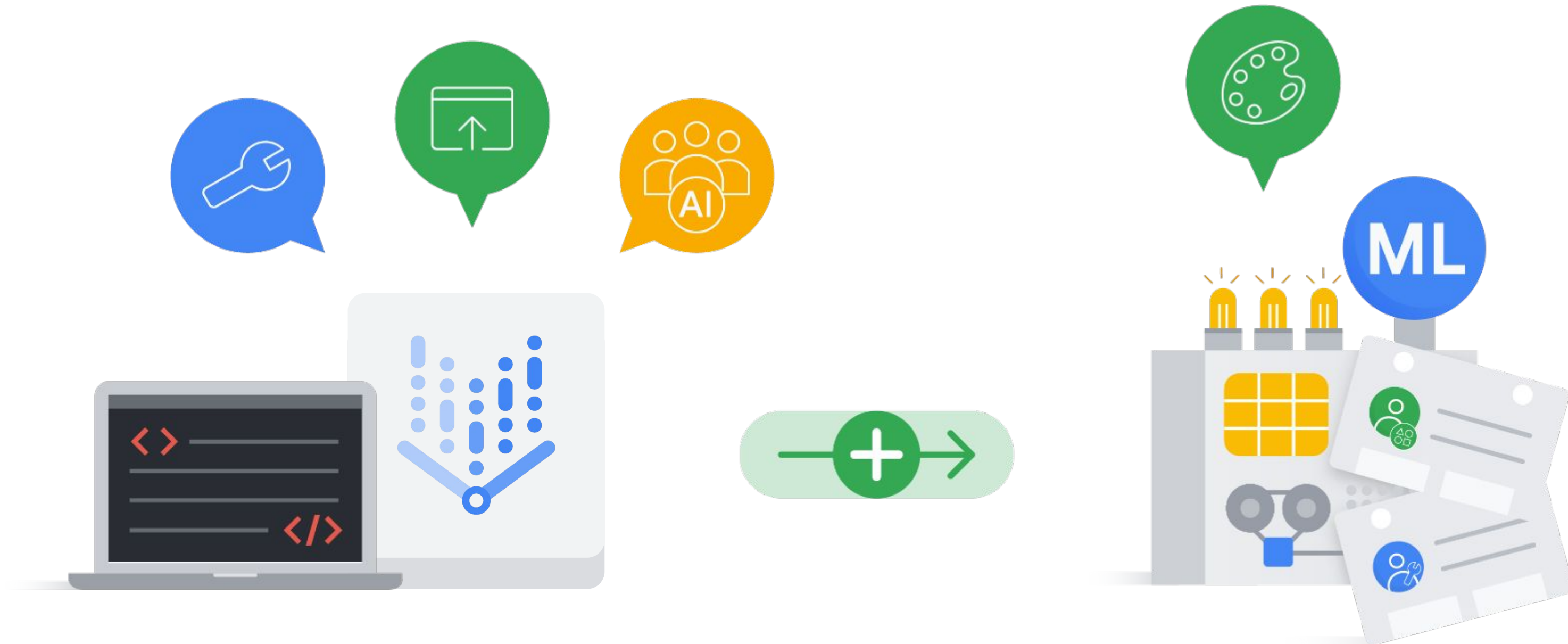


Image recognition

Chat bots

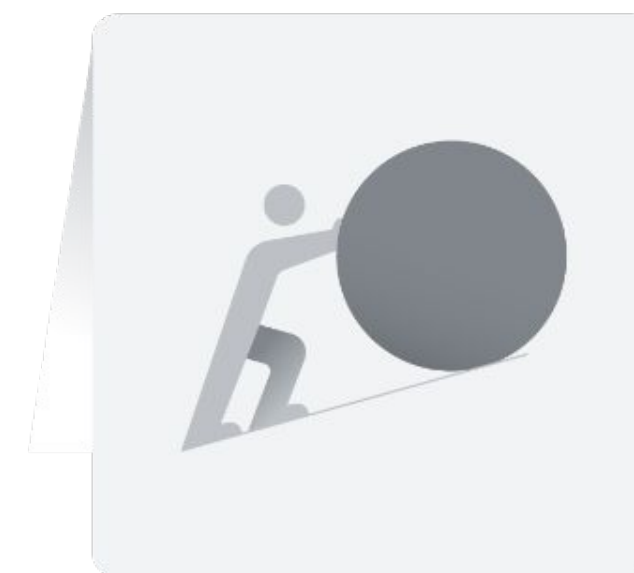
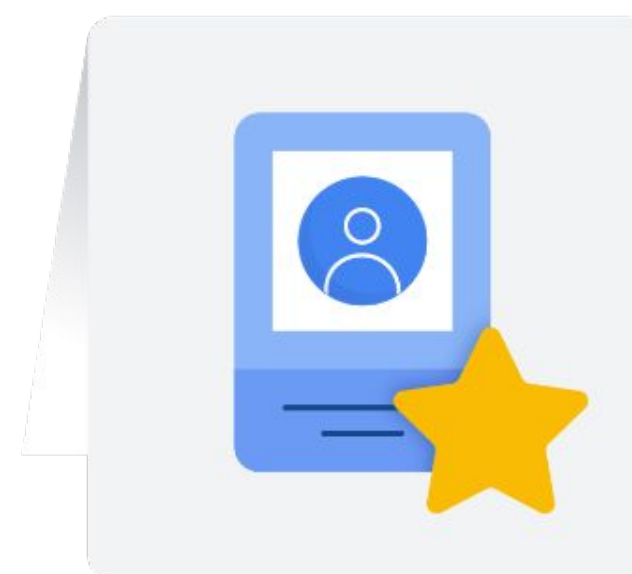
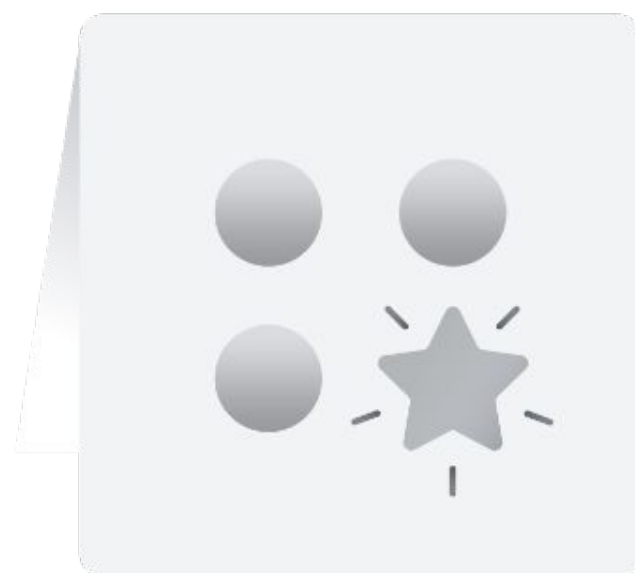
Differentiation



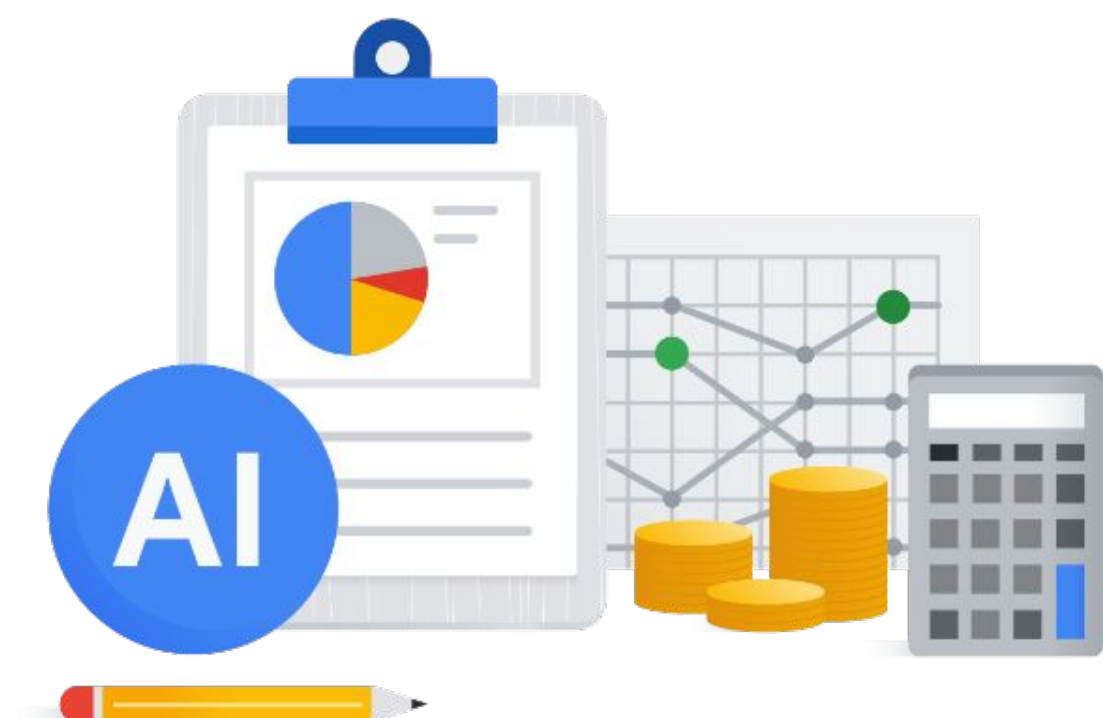
Vertex AI gives ML engineers and data scientists full control of the ML workflow.

What expertise does your project require?

Expertise

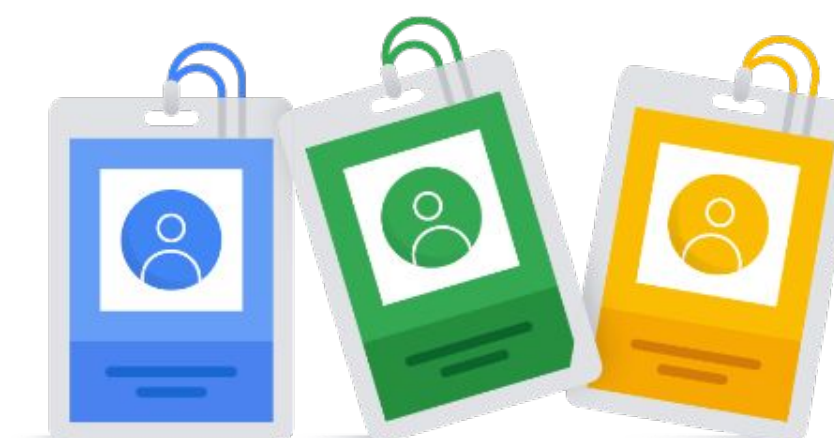


Expertise



Data engineers

Data scientists



Machine learning engineers

Google Cloud's AI and ML products require different types of expertise



Data analysts

Business intelligence teams

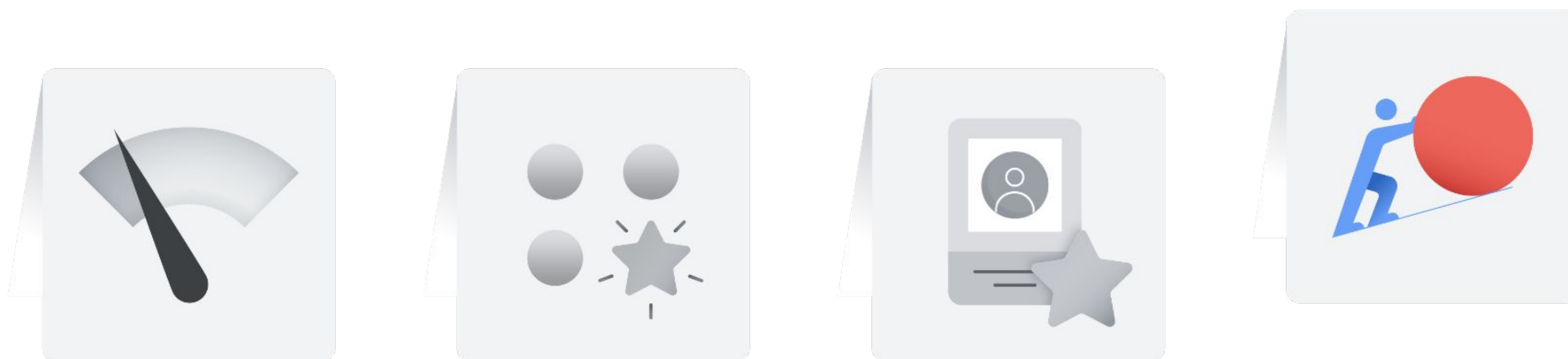


ML engineers

Data scientists

What effort is required to build the AI solution?

Effort



Scenario

Your company is looking to implement an AI solution to improve product recommendations. There's a lot of historical data on customer purchases, but the data is not well structured. In addition to data on customer purchases, your company also has access to customer demographics, product reviews, and social media data.

Considering the tradeoffs of speed, differentiation, expertise, and effort, which of the following options should your company consider?

Pre-trained APIs are quick to deploy and can be used to generate recommendations based on a variety of factors, such as customer purchase history, product popularity, and customer demographics.



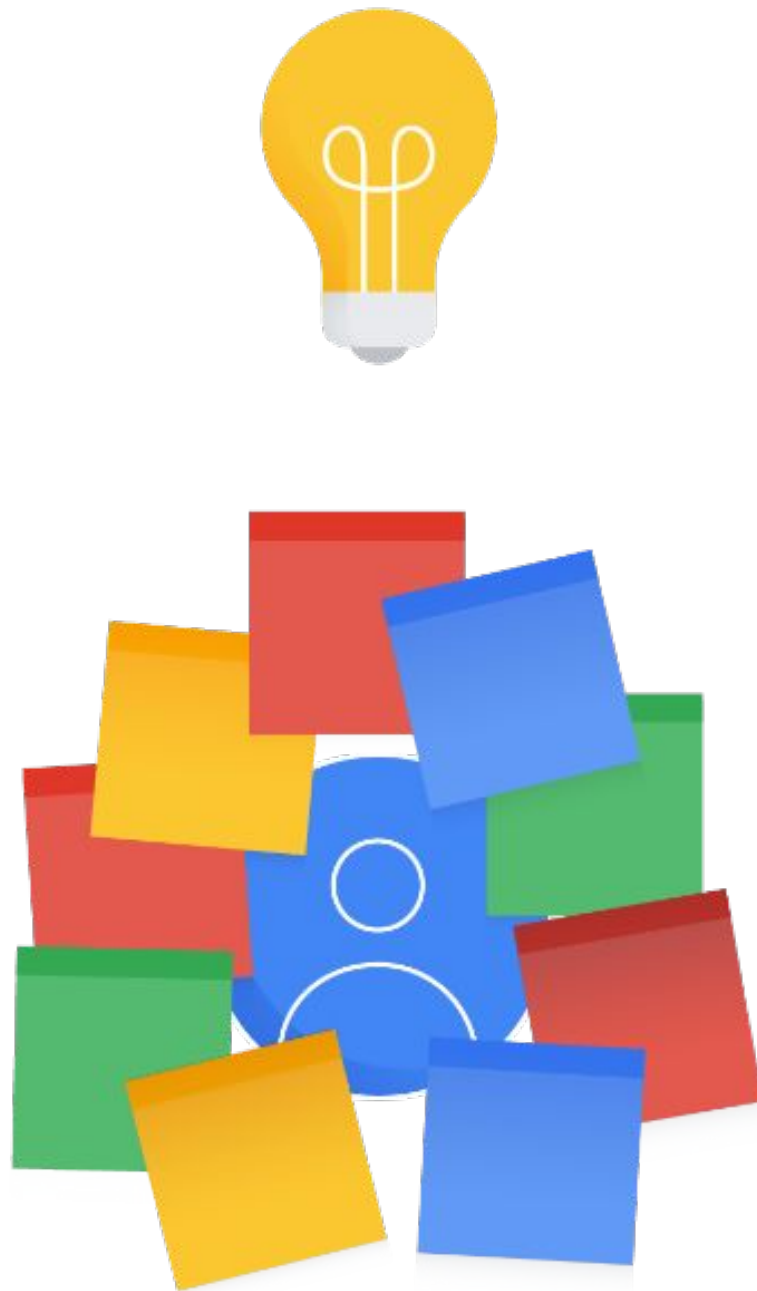
AutoML can be used to build a product recommendation model based on the company's historical purchase data.



Custom training can be used to build a product recommendation model that takes into account all of the company's data, including structured data, unstructured data, and external data sources.



Determining effort depends on several factors



- ✓ The complexity of the problem
- ✓ The amount of data available
- ✓ The experience of the team

Quiz

Question

Artificial intelligence is best suited for replacing or simplifying rule-based systems. Which is an example of this in action?

- A. Implementing AI to develop a new product or service that has never been seen before.
- B. Using AI to replace a human decision-maker in complex situations, such as those involving life-or-death choices.
- C. Using a reinforcement learning algorithm to train autonomous drones for package delivery.
- D. Training a machine learning model to predict a search result ranking.

Quiz

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Quiz

Question

An online retailer wants to help users find specific products faster on their website. One idea is to allow shoppers to upload an image of the product they're looking to purchase. Which of Google's pre-trained APIs could the retailer use to expand this functionality?

- A. Vision API
- B. Natural Language API
- C. Speech-to-Text API
- D. Video Intelligence API

Quiz

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Quiz

Question

Which Google Cloud AI solution is designed to help businesses improve their customer service?

- A. Document AI
- B. Contact Center AI
- C. Discovery AI for Retail
- D. Cloud Talent Solution

Quiz

Answer

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Google Cloud